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Cdhr1a and pcdh15b link photoreceptor outer segments with inner segment calyceal processes revealing a potential mechanism for cone-rod dystrophy

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eLife Assessment

This **valuable** study investigates the interaction of two integral membrane proteins (Cdhr1a and Pcdh15b) and their roles in cone-rod dystrophy. **Convincing** evidence using loss-of-function mutants demonstrates clearly that both proteins are required for cone maintenance and survival. Although some evidence (Western blots and cell aggregation assays) demonstrates Cdhr1a and Pcdh15b can physically interact, there is insufficient evidence to support the subcellular localization and the proposed heterodimeric interaction of the two proteins from distinct subcellular compartments in cone photoreceptors.

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Abstract

Cone rod dystrophy (CRD) is a macular degeneration disorder characterized by initial cone cell photoreceptor degeneration and subsequently of rod photoreceptors. Mutations in CDHR1, a photoreceptor specific cadherin have been found to be associated with the incidence of cone-rod dystrophy and recapitulated in mouse CDHR1 knockouts. However, the molecular function of CDHR1 remains unknown. CDHR1 has been shown to localize at the leading edge of murine rod nascent outer segment (OS) making junctions to an unknown partner in the inner segment. Using Structured Illumination Microscopy (SIM), we observed that the localization of zebrafish *cdhr1a* extends from basal nascent OS discs above the periciliary ridge of the inner segment to a considerable length along the OS, akin to calyceal process (CPs). When labeling the CPs using *pcdh15b*, a CP specific cadherin, we observed that *cdhr1a* at the leading edge of OS juxtaposes with *pcdh15b* in the CP. Similar localization patterns were detected in human, macaque, xenopus, ducks, and various rodent PRCs indicating conservation. Importantly, using immunoprecipitation and K652 cell aggregation assays we demonstrate that *pcdh15b* and *cdhr1a* can interact and potentially link the OS and CP. To analyze the consequences of OS-CP interactions in CRD, we established a zebrafish *cdhr1a* mutant line (*cdhr1a*^{fs*146}) and analyzed CRD progression at high temporal resolution. Homozygous *cdhr1a*^{fs*146} mutants begin to exhibit minor cone OS morphology defects starting at 15 dpf (days post fertilization) and severe OS disruption and cell loss by 3 months. Rod OS defects were delayed until 3-6 months.

Furthermore, we show that loss of *cdhr1a* function leads to disorganization and shortening of CPs coinciding with cone outer OS defects which is significantly exacerbated when combined with the loss of *pcdh15b*. In conclusion, we propose that *cdhr1a* and *pcdh15b* function to link cone OSs with CPs to maintain proper OS homeostasis thus revealing a potential novel mechanism for CRD.

Introduction

Inherited retinal diseases (IRDs) are a group of disorders that can severely affect vision and even lead to blindness affecting over 450,000 individuals in the US alone ¹. Cone Rod dystrophy is a subset of IRD that leads to cone photoreceptor cell (PRC) degeneration followed by decline of rod PRCs ^{1,2}. The disorder generally stems from mutations in one of the over 30 gene candidates. Function of a putative CRD candidate, CDHR1 has been known to clinically correlate with recessive cone rod dystrophy ^{3–5}. CDHR1, a photoreceptor-specific cadherin, was first investigated by Rattner and colleagues in 2001 (previously PCDH21) where it was shown that a mouse KO resulted in early onset retinal degeneration, at one month and severe degeneration by six ⁶. Subsequent studies have shown that initial retinal degeneration involves loss of cones and subsequently rods ⁷. Molecularly, in the mouse model, it's been shown that CDHR1 localizes at the leading edge of OS discs and forms cadherin-based junctions to an unknown partner in the inner segment ^{6,8}. These connections have been hypothesized to regulate nascent rod OS disc release into the mature OS ⁹. However, CDHR1 function in cones has not been postulated. In *Xenopus*, absence of *Cdhr1* results in overgrown OS discs at the basal OS thereby disrupting the normal OS ¹⁰. In humans, CDHR1 mutations, both truncations and missense mutations, have been shown to correlate with late onset CRD ^{5,11–13}. Clinically, these studies identify various pathogenic and variants of unknown significance, however the exact mechanism behind CDHR1 based CRD pathogenesis remains unknown.

CDHR1, like other cadherins encodes an intracellular domain, a transmembrane domain and six evolutionarily conserved extracellular cadherin domain repeats (EC repeats) ^{6,9}. The cytoplasmic domains of classical cadherins have been shown to bind beta catenin, which anchors the cadherin to various cytoskeletal elements such as actin ¹⁴. Interestingly, the cytoplasmic domain of *Cdhr1* diverges considerably from other cadherins and is predicted to not possess any beta catenin or cytoskeletal binding activity ^{9,15}. ECs of most cadherins possess negatively charged sequence motifs, which are involved in Ca^{2+} binding ¹⁶. These ECs form either homophilic or heterophilic junctions in trans conformation and thereby mediate inter/intracellular connections ¹⁷. Protocadherins generally consist of six to seven EC repeats and are predominantly observed with high diversity in various neuronal populations ¹⁸. In neurons, protocadherins are responsible for synaptogenesis, neuronal specificity, and formation of mechanical junctions ¹⁹. Based on mouse studies, CDHR1 forms connections with the IS to what appears to be the periciliary ridge and an extension of the IS appearing to be the calyceal process ⁸.

Calyceal processes (CPs) were first described by Cohen *et al.* in 1961 (as calyx of a flower) surrounding the basal OS discs in rhesus macaque and pigeon photoreceptors ²⁰. Pioneering work in the 1970-80s showed these processes are actin-based microvilli projections of the IS found in both rods and cones of various vertebrates including zebrafish and humans ^{21,22}. In macaque PRCs, thicker, longer and more numerous CPs were observed surrounding cones versus rods ²³. Based on their proximity to the OS, CPs were hypothesized to structurally support the OS discs in both rods and cones ²⁴. However, functional evidence of their potential aid to the OS remains scarce. Previous studies also show how CPs are absent and a vestigial remanence of Periciliary ridge extension is observed in popular rodent models such as mice and rats, which may suggest to the lack of functional evidence of CPs ^{23,25}. Studies in macaque PRCs indicated localization of various Usher syndrome 1 (*Ush1*) proteins to the calyceal processes ²³. Clinically Usher syndrome 1 is categorized by loss of hearing and vision due to mutations in one of the six *Ush1* genes ²⁶. Of these six *Ush1* genes, *Pcdh15* has been shown to make heterophilic cadherin-based junctions in stereocilia of hair cells with another *Ush1* gene, Cadherin 23 (*Cdh23*) ²⁷. Thereby they form a mechanical bridge between adjacent stereocilia.

Mechanical deflection of stereocilia due to sound waves changes the tension of *Pcdh15*-*Cdh23* based interactions, which then triggers the mechano-electrical transduction channel for signal propagation ^{28,29}. Loss of functional *Pcdh15* results in Usher syndrome type 1f, entailing the loss of hearing and vision ³⁰. In PRCs, *Pcdh15* has been unequivocally shown to be localized in the CPs ^{23,31,32}. Interestingly loss of *pcdh15b* function in zebrafish results in an abnormal OS phenotype

and PRC degeneration³¹. However, the mechanism of pcdh15b based PRC pathogenesis remains unknown. Noting the expression of two retinal cadherins in juxtaposing structures, CDHR1 in the OS and PCDH15 in the CP, we sought out to investigate whether CDHR1 and PCDH15 interact to link the OS discs and the CP.

In our current study we demonstrate that in zebrafish PRCs cdhr1a and pcdh15b juxtapose along the OS and CP suggesting they form functional connections. Importantly, we show that this localization pattern is evolutionarily conserved up to and including humans. Furthermore, we confirm pcdh15b-cdhr1a interactions via immunoprecipitation and K562 cell aggregation. While modeling CRD using a cdhr1a loss of function mutant line we report progressive defects in CP morphology correlating specifically with cone degeneration. Taken together, we propose that loss of cdhr1a function disrupts OS-CP junctions and subsequently leads to CRD.

Results

Cdhr1a localization in the OS suggests an interaction with the calyceal process

As previously mentioned, CDHR1 has been shown to localize to the inner/outer segment boundary in several species, and in mouse rod cells it was shown to extend from newly formed discs and form linkages with the inner segment. The connecting partner had yet to be determined. To expand our understanding of CDHR1 localization we performed IHC for cdhr1a and imaged the sections using high resolution confocal microscopy. When focusing on cone cells, we observed that cdhr1a was not restricted to the IS/OS boundary, instead cdhr1a localization was found along the edges of OSs (Fig 1A). Conversely in rod cells we observed that cdhr1a localization was restricted to a much smaller region of the rod OS and closely associated with the IS, more reminiscent of the mouse cryo-EM studies (Fig 1B). The cone pattern of localization was reminiscent of recent work showcasing the actin based IS extensions called calyceal processes (CP). While little is known about the exact molecular function of CPs, recent studies have shown that the Usher syndrome-associated protein pcdh15b, also a protocadherin, localizes to the CP in zebrafish, and in non-human primates. After examining zebrafish retinal expression of 5 key Usher proteins (Pcdh15b, Cdh23, Myo7a, Ush1c, and Ush1g) we only detected pcdh15b in the PRC layer (Fig S1). As expected, pcdh15b localization outlined the presumptive CPs in both cone and rod cells (Fig 1A', B'). Thus, we next directly compared cdhr1a localization to that of pcdh15b and excitedly observed a close association between the two signals in both rods and cones (Fig 1A'', B''). Interestingly, the localization of cdhr1a and pcdh15b did not appear to directly overlap, thus suggesting that the two proteins may juxtapose one another. To examine their localization at higher resolution, we turned to the super resolution technique structured illumination microscopy (SIM). Using SIM, we confidently confirmed that cdhr1a and pcdh15b signals juxtapose along the OS and the CP respectively (Fig 1C). Additionally, we also imaged the localization of actin, which is expected to be restricted to the CP. When comparing cdhr1a and actin signals we observed a juxtaposition rather than complete co-localization (Fig 1D). On the other hand, when comparing pcdh15b and actin we observed near total co-localization suggesting both are localized in the CP (Fig 1E). These localization patterns were visualized in both lateral and transverse views (Fig 1F-H). Taken together, this data suggested that cdhr1a in the OS and pcdh15b in the CP may be interacting.

To further validate cdhr1a localization and the potential for interaction with pcdh15b we employed immunogold-TEM. Using gold-labeled secondary antibodies we were able to co-localize cdhr1a and pcdh15b at the interphase between outer segments and CPs, thus validating our immunohistochemistry results (Fig 1I). Based on these observations, and the fact that both cdhr1a and pcdh15b are from the cadherin family with heterophilic interaction tendencies, we hypothesize that cdhr1a connects the OS to the CP by cadherin-based interactions with pcdh15b (Fig 1J). Furthermore, we predict that cdhr1a-pcdh15b interactions span majority of the outer segment in cones but only connect newly forming discs in rod outer segment. To determine if this hypothesis pertains only to teleost fish, we also examined localization of CDHR1 and PCDH15

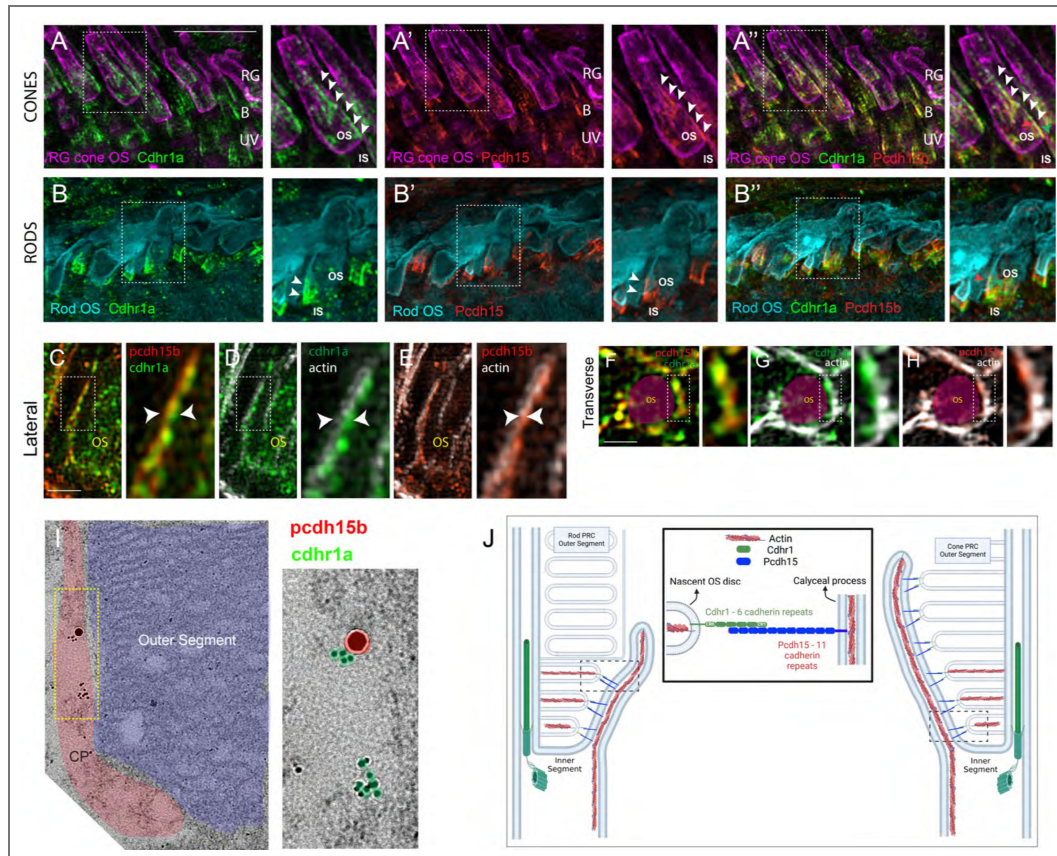


Figure 1. Evolutionarily conserved localization of pcdh15 and cdhr1 in photoreceptors predicts interactions linking outer segments and calyceal processes.

A-A'') Confocal microscopy of wildtype 5dpf retinal cryosections probed with cdhr1a antibody (green), Peanut germ agglutinin (PNA - magenta) to label red-green cones, and pcdh15b (red). White boxes indicate the location of the inset enlargement. White arrowheads highlight the linear localization of cdhr1a along cone OSs (A) and pcdh15b outlining the calyceal process (A'). Merge of all three signals highlights the proximity between cdhr1a and pcdh15b (A''). B=blue cones, RG=red/green cones, UV=UV cones. Scale bar = 5µm. **B-B''**) Confocal microscopy of wildtype 5dpf retinal cryosections probed with cdhr1a antibody (green), pcdh15b (red) and wheat germ agglutinin (WGA -teal) to label rods. White boxes indicate the location of the inset enlargement. White arrowheads highlight the linear localization of cdhr1a along rod OSs (B) and pcdh15b outlining the calyceal process (B'). Merge of all three signals highlights the proximity between cdhr1a and pcdh15b (B''). Scale bar = 10µm. **C-E**) Structured illumination microscopy (SIM) of 5 dpf wildtype zebrafish retinal cryosections probed with **C**) cdhr1a (green) and pcdh15b (red), **D**) pcdh15b (red) and actin (white) and **E**) pcdh15b (red) and actin (white). White boxes represent the inset enlargement. White arrowheads highlight the juxtaposition or overlap between the cdhr1a, pcdh15b and actin signals. Scale bar = 2.5µm. **F-H**) Structured illumination microscopy (SIM) of wildtype zebrafish whole mount 5 dpf retina in transverse orientation (OS outlined in magenta) probed with **F**) cdhr1a (green) and pcdh15b (red), **G**) cdhr1a (green) and actin (white) and **H**) pcdh15b (red) and actin (white). White boxes represent the inset enlargement. Scale bar = 2.5µm. **I**) Immuno-gold-TEM detection of cdhr1a (green) and pcdh15b (red) in 5 dpf wildtype retinal sections. The CP is outlined in red and the OS in blue. Scale bar = 500nm **J**) Diagrammatic model of the connection between the OS discs and CPs in both rod and cone cells mediated by the interaction between OS bound cdhr1a and CP bound pcdh15b.

across various vertebrate species including frog, duck, mouse, rat, gerbil, spiny mouse, macaque and human (Fig 2). In all the species examined we observed a close localization or juxtaposition of *Cdhr1* and *Pcdh15* signals indicating that their association is likely to be evolutionarily conserved. Furthermore, as was observed in zebrafish, in all the species examined the vertical distribution and pattern of *Cdhr1* signal always mirrored that of *Pcdh15*. Taken together, we postulated that the function of *Cdhr1* may be to link the OS discs to the CP and that loss of this interaction could represent a novel mechanism of cone-rod dystrophy.

Cdhr1a and pcdh15b interact and form functional connections

To test our hypothesis that *cdhr1a* and *pcdh15b* physically interact to connect the OS and CP we first assessed their interaction using immunoprecipitation (IP). To do so we cloned *cdhr1a* cDNA fused to a C-terminal FLAG tag into a mammalian expression vector and *pcdh15b* fused to a C-terminal MYC tag. Using HEK293 cells we performed co-transfection and subsequently FLAG and MYC pull down assays. Our results show that *cdhr1a* and *pcdh15b* can reciprocally pull each other down in the IP assay (Fig 3A). While IP can indicate interaction between proteins or between complexes of proteins, our hypothesis predicts that the interactions between *cdhr1a* and *pcdh15b* are functional in that they connect the OS and CP in extracellular space. To test whether *cdhr1a* and *pcdh15b* can form functional extracellular complexes we used the K562 leukemia cell line, which lacks any endogenous cadherin protein expression and therefore does not form cell aggregates. Based on this fact, K562 cells are often used to test cadherin-cadherin interactions³³. As such, we transfected K562 cells with either *cdhr1a*-FLAG or *pcdh15b*-MYC to test whether their interaction can result in K562 aggregation (Fig 3B). Introduction of either cadherin alone resulted in minimal aggregation, either in total number of aggregates or the number of cells per aggregate (Fig 3C-E). Conversely, when we mixed the *cdhr1a*-FLAG and *pcdh15b*-MYC expressing cultures we observed a significant increase in the total number of aggregates and the number of cells in each aggregate (Fig 3C-E). Based on these results we conclude that *cdhr1a* and *pcdh15b* can form cadherin-based junctions in extracellular space and therefore have the potential to establish connections between the OS and CP in the retina.

Generation of a zebrafish *cdhr1a* loss of function line for the study of cone-rod dystrophy

CDHR1 deficiency has been modelled using the murine model which recapitulates the human phenotypes associated with cone-rod dystrophy³⁴. However, despite two decades since the model was first generated, the mechanism behind PRC degeneration has yet to be elucidated. While recapitulating the disease phenotypes, the mouse model does exhibit some limitations, including having a rod-rich retina as well as the logistical difficulty in examining numerous timepoints during early development, adolescence and throughout adulthood. As such, we sought to model CDHR1-mediated cone-rod dystrophy using zebrafish which offers a vertebrate cone-rich model that readily enables detailed molecular examination of PRCs at high temporal resolution. Thanks to high conservation of ocular development, genetics and function, the zebrafish model has been used to model numerous human ocular diseases including retinitis pigmentosa, LCA, coloboma and numerous others.

Zebrafish encode two CDHR1 homologues, *cdhr1a* and *cdhr1b*, of which only *cdhr1a* is expressed in the zebrafish photoreceptors³⁵. We targeted *cdhr1a* using the Alt-R CRISPR technology by designing two crRNA constructs separated by ~170bp and mapping to exon 6 and exon 7 (Fig 4A). Upon injection, we generated a stable zebrafish line harboring a 173bp deletion which results in the excision of the 3' end of exon 6, all of intron 6 and most of exon 7 ultimately generating a premature stop codon at AA146 (Fig 4B,C). *Cdhr1a* protein sequence encodes 6 extracellular cadherin domains (EC1-6), a transmembrane domain and a cytoplasmic domain (Fig 4D). The 173bp deletion is predicted to result in a truncation of this polypeptide at AA146, therefore terminating the protein after just the first EC domain. Thus, the remaining protein would lack both the transmembrane and cytoplasmic domain and therefore likely lose all function. The *cdhr1a*^{fs146*} line (from here on referred to as *cdhr1a*^{-/-}) was bred to homozygosity to

Figure 2. Evolutionarily conserved localization of pcdh15 and cdhr1 in photoreceptors predicts interactions linking outer segments and calyceal processes.

Structured illumination microscopy (SIM) of wildtype xenopus (A), mallard duck (B), macaque (C), human (D), mouse (E), rat (F), spiny mouse (G) and gerbil (H), retinal sections probed for cdhr1 (green), pcdh15 (red) and PNA (magenta) to label cone outer segments. White boxes represent the inset enlargements. White arrowheads highlight the juxtaposition of the cdhr1a and pcdh15b signals in each species. Scale bar = 5µm.

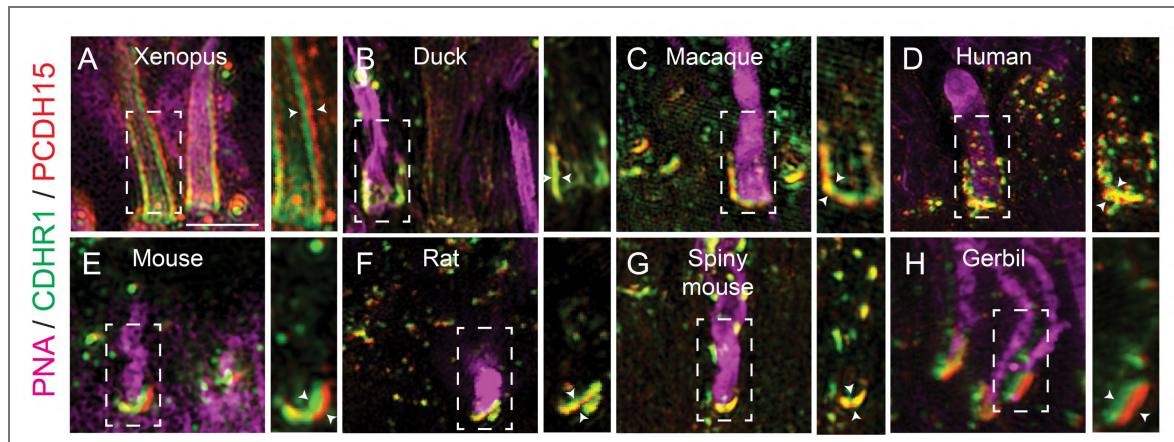
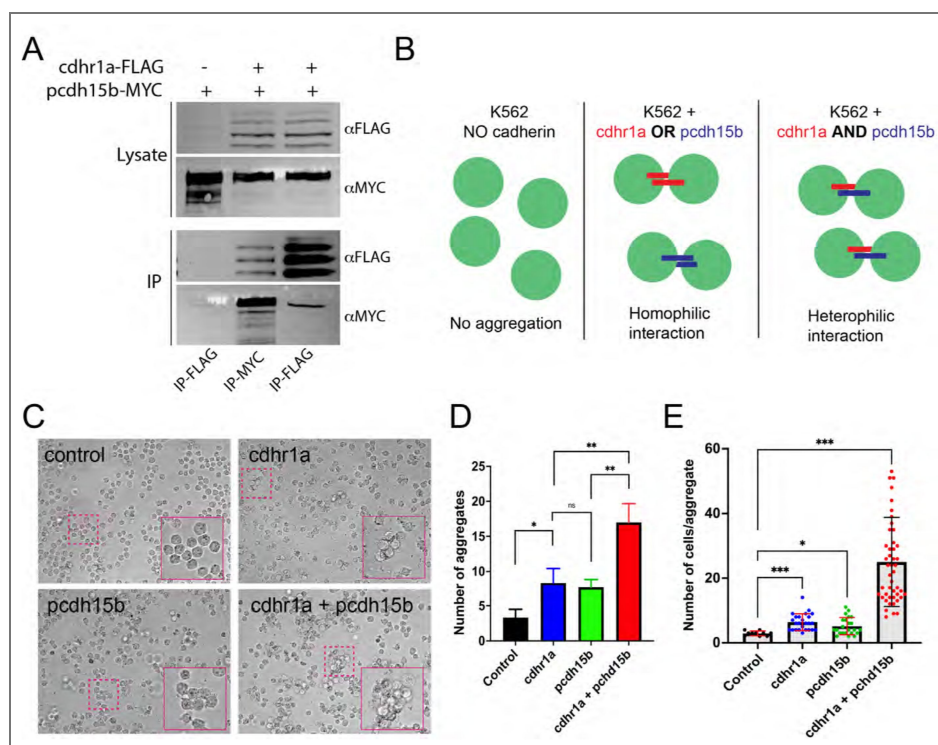


Figure 3. Physical interactions between cdhr1a and pcdh15b can facilitate cell-cell adhesion.

A) Immunoprecipitation of cdhr1a-FLAG and pcdh15b-MYC performed in HEK293 cell lysates. Pull down of cdhr1a-FLAG using anti-FLAG antibody beads also pulls down pcdh15b-MYC, while pull down of pcdh15b-MYC using anti-MYC antibody beads also pulls down cdhr1a-FLAG. **B)** Diagrammatic representation of the K562 cell assay for assessing homophilic or heterophilic interactions between cadherins. **C)** Brightfield microscopy analysis of K562 cell aggregation after transfection with either cdhr1a or pcdh15b or after co-culture of the two transfected populations. Magenta boxes indicate regions enlarged in the insets. **D)** Quantification of the total number of aggregates observed in a single field of view. **E)** Quantification of the number of average number of cells in each observed aggregate.



create a maternal zygotic mutant population. The maternal zygotic mutants were used for all the subsequent functional experiments. To confirm that our mutant line represented a null allele we probed for *cdhr1a* in PRCs using a zebrafish specific *cdhr1a* antibody which we previously developed and verified ³⁵. As expected, immunohistochemistry (IHC) of 5 dpf wildtype zebrafish retinal cryosections displayed strong *cdhr1a* staining at the periphery of photoreceptor outer segments (Fig 4E [↗](#)). However, *cdhr1a*^{-/-} samples exhibited a total lack of *cdhr1a* protein signal, therefore confirming that our mutant line harbors a null allele. Additionally, *in situ* hybridization data suggests that the allele undergoes non-sense mediated decay and we do not detect any retinal expression of *cdhr1b* as compensation (Fig S2 [↗](#)). Taken together we were able to generate a novel zebrafish line harboring a heritable null allele in *cdhr1a* and establish a homozygous population of fish suitable for functional analysis of PRC phenotypes throughout development, adolescence and adulthood.

Loss of *cdhr1a* function in zebrafish results in progressive cone-rod dystrophy

There are no studies to date that have examined effects of CDHR1 loss of function on CPs. As such, to provide a more comprehensive examination of PRC phenotype progression we examined 7 timepoints encompassing early development (5 and 15 dpf), juveniles (30 dpf) as well as adults (90, 180, 360 and 720 dpf). At each timepoint we first examined the morphology and number of cone cells as it is expected that these cells will be affected first. To visualize cone cell OSs, we used a peripherin2 (*prph2*) antibody which labels the periphery of OSs in PRCs, including all the cone subtypes. We analyzed five individuals for each genotype and at each time point. All COS in the field of view were measured which on average included 40 cone cells at early timepoints (5-15dpf) and 70+ cone cells at later stages (30+ dpf). For consistency, we chose to sample all our timepoints at the central retina (adjacent to the optic nerve). Starting at 5 dpf we observed that both WT and *cdhr1a*^{-/-} cone OSs exhibited their expected shape (Fig 5A, A' [↗](#)) and displayed smooth vertical peripheral *prph2* staining. To determine whether there are any physiological changes to the cone OS (COS) we also measured the OS length based on *prph2* signal. When comparing WT to *cdhr1a*^{-/-} mutants we observed a slight, but significant increase in the average COS size from 4.79μm to 5.34μm (Fig 5H, I [↗](#)). At 15 dpf we again detected an increase in the average COS length in the mutants from 6.74μm in WT to 7.95μm (Fig 5H, I [↗](#)).

We also began to observe that the COS shape appeared to become disorganized, perhaps due to the increase in length (Fig 5B, B' [↗](#)). Interestingly, by 30dpf, when the retina has been fully developed, we observed a significant decrease in average COS length in the mutants, from 7.63μm in WT to 6.68μm (Fig 5H, I [↗](#)). Additionally, the COS of the mutants continued to appear disorganized, lacking the smooth vertical *prph2* labeling observed in WT (Fig 5C [↗](#)) and instead displaying an almost disc like organization (Fig 5C [↗](#)). At 90 dpf there continued to be a decrease in average COS length compared to WT, 8.40μm vs 6.81μm (Fig 5H, I [↗](#)) and increased OS disorganization. Furthermore, at 90 dpf we noted a significant decrease in the number of cones suggesting that there was degeneration of cone cells (Fig 5D, D', J [↗](#)). The most severe phenotypes were observed at the later timepoints. At 180 dpf COS length further decreased to 4.84μm while the wildtype remained at 8.62μm (Fig 5H, I [↗](#)). Furthermore, there was a striking reduction in the number of cone cells remaining (Fig 5J [↗](#)) while COS morphology exhibited severely stunted and tilted COSs (Fig 5E [↗](#)). Analysis of 360 and 720 dpf timepoints displayed similar results to 180 dpf indicating that a maximum level of cone outer segment degeneration and cone loss is achieved by 180 dpf (Fig 5F-G [↗](#)). Taken together, our examination of COS morphology, size and number of cone cells indicates that in the absence of *cdhr1a* function cone cells lose normal OS morphology which precedes their degeneration, which begins between one and three months.

In addition to fluorescence confocal microscopy analysis of OS morphology, we also examined COSs at electron microscopy resolution using TEM. When comparing WT to *cdhr1a* mutants at 15 dpf (n=5) we can already detect mis organization of the OS discs in the form of bulges and improperly stacked discs (Fig 5K-L [↗](#)). This result corresponds with our fluorescence microscopy data (Fig 5B [↗](#)) where at 15 dpf we already see alteration in the pattern of *prph2* staining.

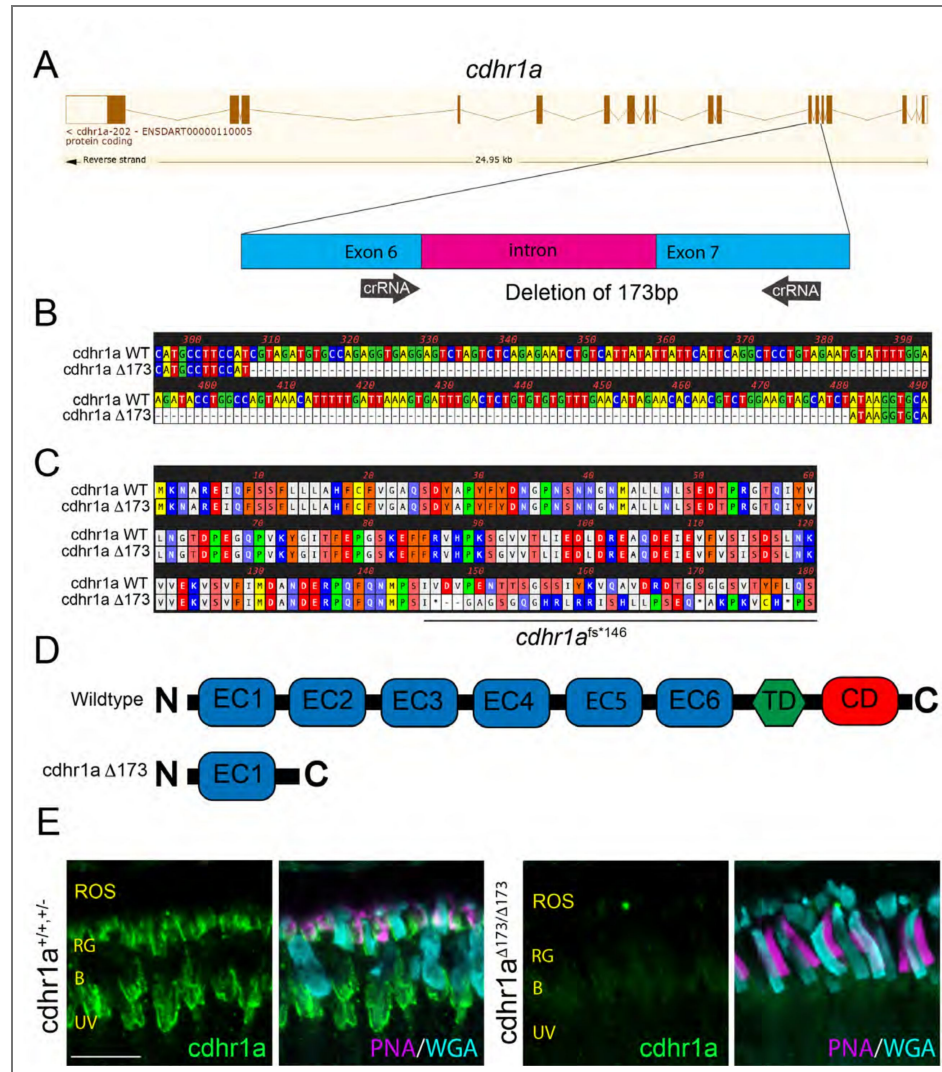


Figure 4. Construction and confirmation of the *cdhr1a*^{-/-} line.

A) Exon/intron diagram of *cdhr1a*. The intron 6-exon 6-intron 7 junction is highlighted and the approximate location of the crRNA (black arrows) is depicted. **B)** Genomic nucleotide sequence from the *cdhr1a*^{Δ173} line highlighting the sequence location of the 173 bp deletion compared to WT sequence. **C)** Amino acid sequence alignment between wildtype and the *cdhr1a*^{Δ173} line. Alignment indicates that the 173bp deletion leads to a frameshift and an immediate premature stop codon at AA146. **D)** Diagrammatic representation of the domain structure of WT *cdhr1a* protein vs the *cdhr1a*^{Δ173} allele. EC = cadherin domain, TD = transmembrane domain, CD = cytoplasmic domain. **E)** Confocal microscopy of 5 dpf retinal cryosections from WT and *cdhr1a*^{-/-} retinas stained with anti-*cdhr1a* antibody (green), PNA (red-green cones - magenta) and WGA (blue cones - teal). *Cdhr1a* signal is detected along photoreceptor outer segments in the wildtype and completely missing in *cdhr1a*^{-/-}. ROS = rod outer segment, RIS = rod inner segment. Scale bar = 5μm.

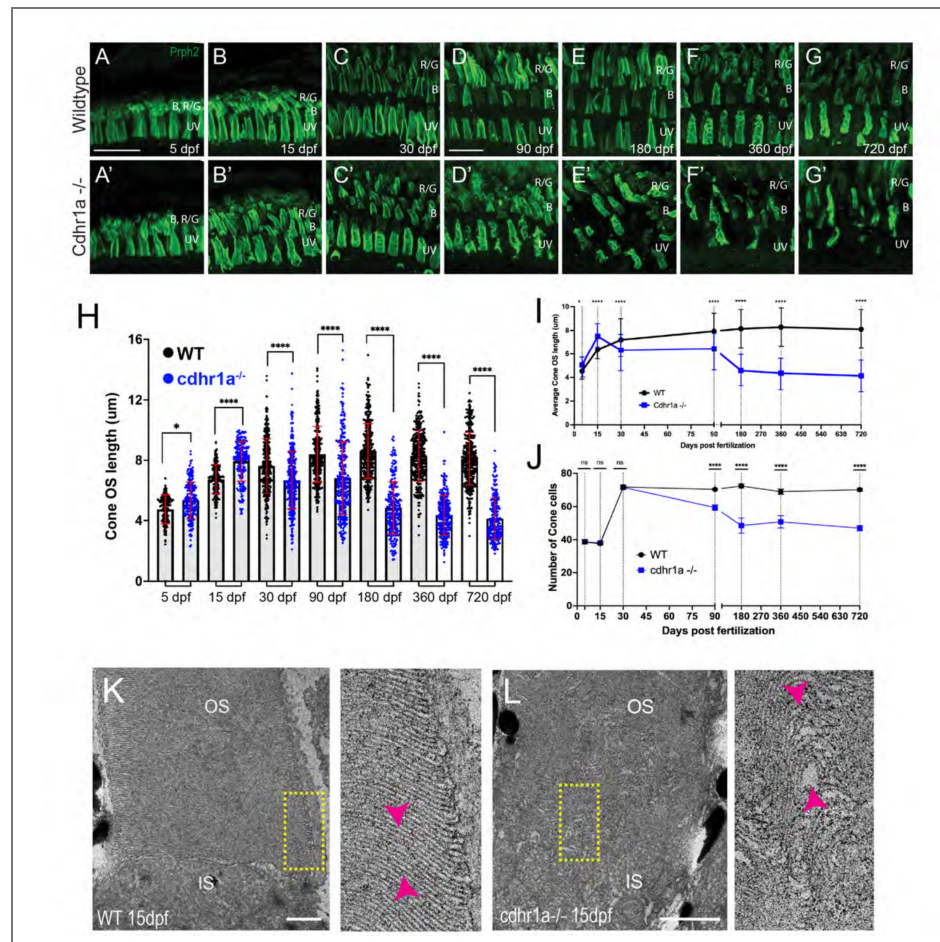


Figure 5. Loss of *cdhr1a* function leads to cone outer segment degeneration.

A-G) Confocal microscopy of wildtype retinal cryosections probed with anti-prph2 antibody (green). B=blue cones, RG=red/green cones, UV=UV cones. Scale bar = 10μm. **A'-G')** Confocal microscopy of *cdhr1a*^{-/-} retinal cryosections probed with anti-prph2 antibody (green). A-B' scale bar = 5μm, C-G' scale bar = 10μm. **H)** Quantification of cone OS length at 5, 15, 30, 90, 180, 360 and 720 dpf measured as length of prph2 signal in wildtype (black dots) and *cdhr1a*^{-/-} (blue dots). Standard deviation is shown in red. **** = $p < 0.0001$. ANOVA = $p < 0.0001$. **I)** Line graph depicting the long-term trend of cone OS length changes between wildtype (black) and *cdhr1a*^{-/-} (blue). **J)** Line graph depicting changes in the number of cone cells counted in the observation region over time in wildtype (black) compared to *cdhr1a*^{-/-} (blue). **K)** Transmission electron microscopy micrographs from a 15dpf wildtype retina. Yellow rectangle represents the enlarged inset. Magenta arrow heads highlight the proper stacking of OS discs. IS= inner segment, OS = outer segment. Scale bar = 500nm. **L)** Transmission electron microscopy micrographs from a 15dpf *cdhr1a*^{-/-} retina. Yellow rectangle represents the enlarged inset. Magenta arrow heads highlight the improper and distorted stacking of OS discs. IS= inner segment, OS = outer segment. Scale bar = 500nm.

Compared to recent work in frogs, we did not observe obvious overgrowth of discs¹⁰. This may be due to developmental differences between frogs, which undergo metamorphosis, and zebrafish. It may also represent differences in the timepoints analyzed as it remains possible that overgrowth in zebrafish, which exhibits rapid development, may be observed at earlier stages. Taken together, our data indicates that COS integrity is significantly affected by the loss of *cdhr1a* as early as 15 dpf, and that this results in progressive loss of cone cells starting at 30 dpf, accelerating up to 180 dpf and stabilizing from 360 to 720 dpf (Fig 5J [↗](#)).

While cone cells were the primary target of our investigation of cone-rod dystrophy in our *cdhr1a*^{-/-} model, we also examined the fate of rod cells. Like our analysis of cones, we sought to examine rod OS morphology, length and number of rod cells at the aforementioned timepoints. Due to a delay in the assembly of the rod OSs (ROS) compared to COSs we used Gnb1 antibody (Δ subunit of rod specific transducin) to identify rod cells and wheat germ agglutinin (WGA) to visualize their OS in early development (5 and 15 dpf), and subsequently prph2 and WGA at later stages (30-720 dpf). We analyzed five individuals for each genotype and at each time point. All ROS in the field of view were measured and on average we counted 40 rod cells in each field of view. When analyzing both the morphology and length of ROSs we observed that WT and *cdhr1a*^{-/-} ROS were similar in appearance and number at early timepoints (Fig 6A-C', H-J [↗](#)). Interestingly, unlike COSs, ROSs did not exhibit any significant increase in length, in the *cdhr1a* mutants at 15dpf and 30dpf (Fig 6H-I [↗](#)). The first phenotypic difference for ROS was observed at 90dpf, where we detected a slight decrease in the average length, 26.6μm vs 30.3μm in WT (Fig 6H [↗](#), I). Furthermore we also noticed the first examples of morphological disorganization (Fig 6D, D' [↗](#)) as well as a 3.4% decrease in the number of rod cells (Fig 6J [↗](#)). The trend of shorter ROS and disorganized morphology continued up to 180dpf where the average OS length decreased to 20.4μm vs 33.7μm for WT. Furthermore, at 180 dpf the number of rod cells decreased by 10%, likely stemming from the persistence of disorganized OSs leading to rod cell degeneration (Fig 6E, E', H-I [↗](#)). Similar to what we observed with cones, rod phenotypes were not further exacerbated at either 360 or 720 dpf (Fig F-G', H-J). Taken together, we conclude that rod cells are only marginally affected by the loss of *cdhr1a* function in early development and early adulthood and show significant phenotypic effects only starting at 180 dpf. Based on our data, we predict that the molecular mechanism driving cone-rod dystrophy in our mutant line is primarily involved with the maintenance of COS homeostasis. Rod effects are either secondary responses to the loss of cones or due to differences in the function of *cdhr1a* between rods and cones.

Loss of *cdhr1a* function leads to CP disorganization

Since our hypothesis involves connections between OSs and CPs, we next examined the consequence of *cdhr1a* function on CPs. Several previous studies have established that zebrafish exhibit robust numbers of CPs whose assembly coincides with formation of the COSs and ROSs^{24,31}. To examine CP integrity, we measured their length and observed overall morphology using actin antibody staining which is commonly used to assess CP size and morphology^{23,24,31,32}. We measured every rod and cone CP observed in the field of view from five individuals at each time point. To assign the correct identity of the CP and correctly measure its length, counter stains to identify cones (PNA) vs rods (gnb1 or WGA) and retinal location using prph2 staining were used. CP measurements were done using prph2 counter staining as to delineate the base of the OS as the starting point of the CP (Fig 7 Supplement). Imaging was restricted to sections collected from our standardized anatomical location which included the lens. In line with previous results, we observed a gradual increase in CP length in both cones and rods throughout early development (5-15 dpf) and a steady state from adolescence to adulthood (30-180dpf) (Fig 7Q, R [↗](#)). Furthermore, we observed that zebrafish cone CPs are significantly longer, more than double, than their rod counterparts at all time points analyzed. Longer cone CPs were previously also observed in macaque retina²³. Next, we compared wildtype CPs to those of the *cdhr1a*^{-/-} mutants. At 5 and 15 dpf we observed that morphologically cone CPs in *cdhr1a*^{-/-} resembled those of WT (Fig 7A-B [↗](#)). However, even at 5 dpf measurements of CP length indicated a slight decrease in CP size in the mutants, 3.70μm vs 3.49μm (Fig 7O [↗](#)) while at 15 dpf mutant CPs measured longer than wildtype, 5.54μm vs 6.72μm (Fig 7B, B' [↗](#)). Similar results were observed for rods where mutant CPs were

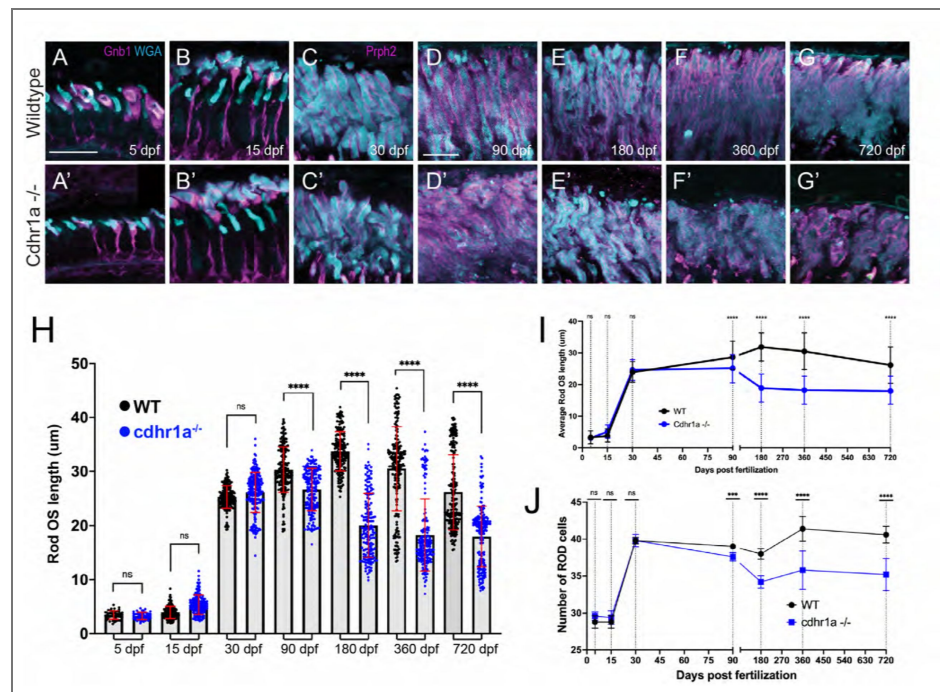


Figure 6. Loss of *cdhr1a* function leads to late onset rod outer segment degeneration.

A-B) Confocal microscopy of wildtype retinal cryosections probed with Gnb1 antibody (magenta) and WGA (teal) to identify rod outer segments at 5 and 15 dpf. Scale bar = 10μm. **A'-B')** Confocal microscopy of *cdhr1a*^{-/-} retinal cryosections probed with Gnb1 antibody (magenta) and WGA (teal) to identify rod outer segments at 5 and 15 dpf. Scale bar = 10μm. **C-G)** Confocal microscopy of wildtype retinal cryosections probed with prph2 antibody (magenta) and WGA (teal) to identify rod outer segments. A-B' scale bar = 5μm, C-G' scale bar = 10μm. **C'-G')** Confocal microscopy of *cdhr1a*^{-/-} retinal cryosections probed with prph2 antibody (magenta) and WGA (teal) to identify rod outer segments. Scale bar = 10μm. **H)** Quantification of rod OS length at 5, 15, 30, 90, 180, 360 and 720 dpf measured as length of WGA signal in wildtype (black dots) and *cdhr1a*^{-/-} (blue dots). Standard deviation is shown in red. ** = $p < 0.001$, **** = $p < 0.0001$. ANOVA = $p < 0.0001$. **I)** Line graph depicting the long-term trend of rod OS length changes between wildtype (black) and *cdhr1a*^{-/-} (blue). **J)** Line graph depicting changes in the number of rod cells counted in the observation region over time in wildtype (black) compared to *cdhr1a*^{-/-} (blue).

longer at 15 dpf (Fig 7I-I', P). Starting at 30dpf we began to observe not only morphological disorganization of actin staining along CPs, but also a significant decrease in overall CP length in *cdhr1a*^{-/-} samples (Fig 7C,C'). Cone CPs appear to have lost their tether to the OS and tilt away from the OS. Interestingly, *cdhr1a*^{-/-} rod CPs do not show any effects in their morphology or length at 30 dpf (Fig 7H, H'). This trend continued and intensified in cone CPs at the 90 and 180 dpf timepoints (Fig 5D-E', F'). In fact, by 180 dpf we observed the length of cone CPs shorten to what it was at 5dpf, which is half of what WT exhibits at 180 dpf, 7.36μm vs 3.47μm (Fig 5E, E', F').

Like we observed for COS length, later timepoints of 360 or 720 dpf did not demonstrate further deterioration of the cone CPs (Fig 7F-G', O, Q). For rod CPs, we observed a much subtler effect, where the CP length isn't affected until 90dpf, and even at 180 dpf the degree of decrease in size, 2.67μm vs 2.39μm, while significant was not relatively as large compared to cone CPs (Fig 7K-L', P). Similar to cone CPs, later timepoints did not exhibit any further decrease in rod CP length or organization (Fig 7M-N', R). Having noted significant effects on CP size and morphology in the absence of *cdhr1a* function, we also examined for consequences on *pcdh15b* localization. At 5dpf, *cdhr1a*^{-/-} cone and rod cells displayed wildtype like localization of *pcdh15b* (Fig 7O). However, *pcdh15b* localization in subsequent timepoints mirrored the disorganization of the CP previously observed using actin (Fig 7A-E'). By 180 dpf, very little *pcdh15b* signal remains (Fig 7J) suggesting significant disorganization of CPs but not its complete disassembly. Based on our actin CP staining as well as *pcdh15b* signal, we conclude that at early timepoints (5-90dpf) the absence of *cdhr1a* does not affect the assembly of the CP nor localization of *pcdh15b* to the CP, however, long term (180dpf +) absence of *cdhr1a* correlates with the eventual loss of *pcdh15b* signal from the CPs and disorganization of its structure as observed by actin staining.

Pcdh15b is necessary for proper localization of *cdhr1a*

Recent studies in zebrafish have indicated that loss of *pcdh15b* results in malformation of OSs and PRC degeneration³¹. To examine how the loss of *pcdh15b* compares to the loss of *cdhr1a* we generated a CRISPR/cas9 knockout line by targeting two crRNAs in exon 5, which resulted in a heritable 68bp deletion (Fig 8A). The consequence of the deletion was a frame shift at AA117 and a premature stop at AA118. Like our *cdhr1a*^{fs*146} allele, the *pcdh15b*^{fs117*118} allele (subsequently referred to as *pcdh15b*^{-/-}) results in a truncation of the protein before either the transmembrane domain or the cytoplasmic domain, thus likely representing a null. To confirm the null phenotype, we probed for *pcdh15b* in retinal cryosections using IHC and observed a lack of *pcdh15b* signal in *pcdh15b*^{-/-} samples, validating the allele as a null (Fig 8B-B'). Due to critical function of *pcdh15b* in the inner ear, *pcdh15b*^{-/-} larva were only viable up to approximately 10 dpf, an outcome previously reported in zebrafish³¹. To observe the effects of *pcdh15b* loss of function on PRCs we first analyzed cone OS morphology and size. Due to the immaturity of rods at 5 and 10 dpf and cone specific effects observed in the *cdhr1a*^{-/-} mutant, we focused our attention to cone cells. Using *prph2* staining we observed little effect on COS morphology or length at 5dpf (Fig 8C-C'). However, at 10 dpf *pcdh15b*^{-/-} COSs displayed disorganization and a slight decrease in COS length (Fig 8D-D', E). Next, we examined CP morphology, where at 5 dpf CP morphology in mutants resembled that of wildtype (Fig 8F-F'), however, by 10 dpf mutant CPs showed signs of disorganization (Fig 8G-G'). Quantification of CP length at 5 dpf *pcdh15b*^{-/-} larva revealed a slight, yet significant increase in CP length although by 10dpf no difference in CP length between wildtype and mutant was detected (Fig 8H). Lastly, we examined whether loss of *pcdh15b* had any effects on *cdhr1a* localization. To quantify we measured the length of *cdhr1a* signal along the OS. In doing so we observed that loss of *pcdh15b* did not preclude *cdhr1a* from localizing to OS, however as early as 5dpf *pcdh15b* mutants display a significant decrease in the spread of *cdhr1a* signal along the OS (Fig 8I-I'), which became much more pronounced by 10 dpf (Fig 8J-J'). Quantification of *cdhr1a* signal length along the OS revealed a significant decrease as early as 5 dpf, 3.22μm vs 2.59μm, and a major decrease by 10 dpf, 4.06μm vs 1.64μm (Fig 8K). Taken together we conclude that while *pcdh15b* is not required for *cdhr1a* localization to the OS, spread of *cdhr1a* signal along the cone OS, which likely represents OS-CP interactions, requires *pcdh15b*.

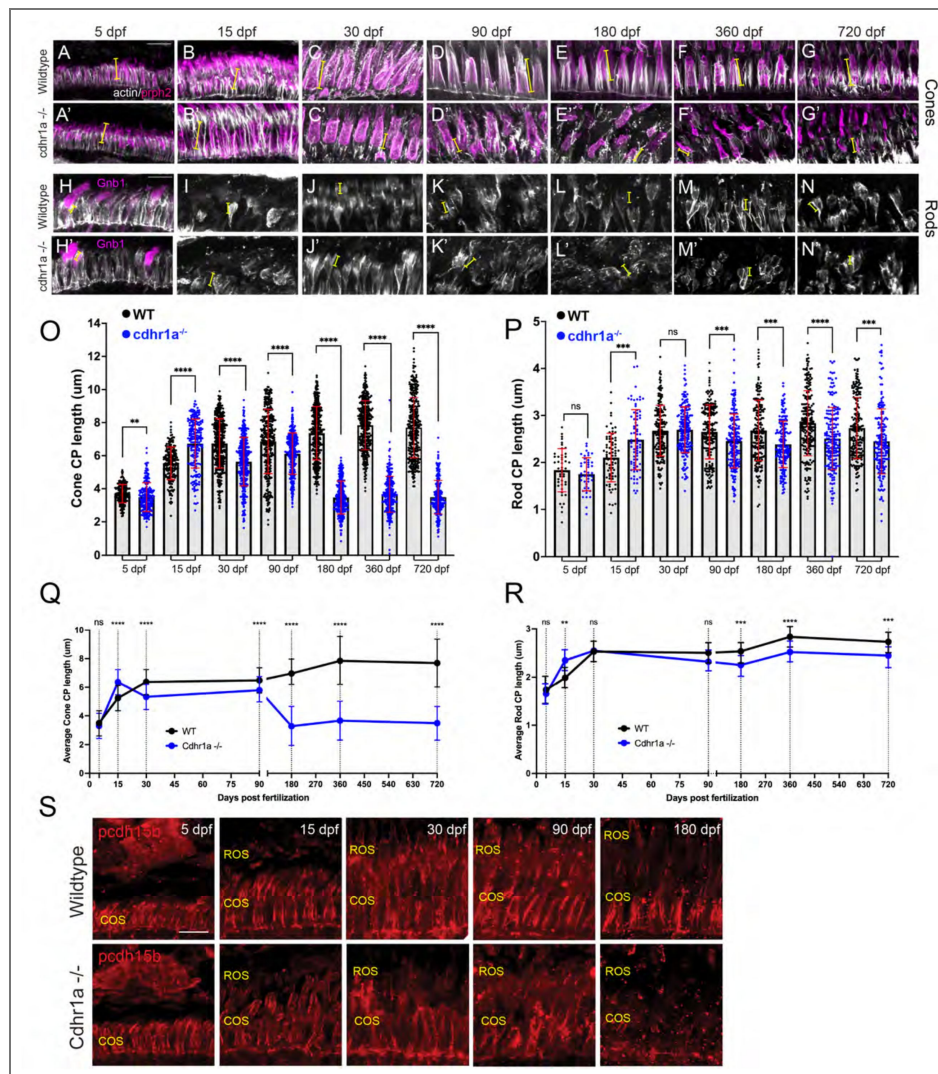


Figure 7. Calyceal processes in cones are disorganized in the absence of *cdhr1a* function.

A-G) Confocal microscopy of wildtype retinal cryosections probed with PNA (magenta) and actin-antibodies (white) at various timepoints detecting cone CPs. Typical length of CPs observed is demonstrated by yellow lines. Scale bar = 5µm. **A'-G')** Confocal microscopy of *cdhr1a*^{-/-} retinal cryosections probed with actin-antibodies (white) at various timepoints detecting cone CPs. Typical length of CPs observed is demonstrated in magenta. **H-N)** Confocal microscopy of wildtype retinal cryosections probed with actin-antibodies (white) and gnb1 (A') (magenta) at various timepoints detecting rod CPs. Typical length of CPs observed is demonstrated in yellow. **H'-N')** Confocal microscopy of *cdhr1a*^{-/-} retinal cryosections probed with actin-antibodies (white) and gnb1 (F') (magenta) at various timepoints detecting rod CPs. Typical length of CPs observed is demonstrated by yellow lines. Scale bar = 5µm. ROS = cone outer segment layer. **O)** Quantification of cone cells CP length based on measurements of actin staining in wildtype (black) compared to *cdhr1a*^{-/-} (blue) at 5, 15, 30, 90, 180, 360 and 720 dpf. Standard deviation is shown in red. ** = p<0.01, **** = p<0.0001. **P)** Quantification of rod cells CP length based on measurements of actin staining in wildtype (black) compared to *cdhr1a*^{-/-} (blue) at 5, 15, 30, 90, 180, 360 and 720 dpf. Standard deviation is shown in red. ns = not significant, *** = p<0.001, **** = p<0.0001. **Q)** Line graph depicting the long-term trend of cone cells CP length changes between wildtype (black) and *cdhr1a*^{-/-} (blue) over time. ** = p<0.01, **** = p<0.0001. **R)** Line graph depicting the long-term trend of rod cells CP length changes between wildtype (black) and *cdhr1a*^{-/-} (blue) over time. ns = not significant, *** = p<0.001, **** = p<0.0001. **S)** Confocal microscopy of wildtype (top panels) and *cdhr1a*^{-/-} (bottom panels) retinal cryosections probed with pcdh15b antibody (red) at various time points. Scale bar = 5µm. COS = cone outer segment layer, ROS = rod outer segment layer.

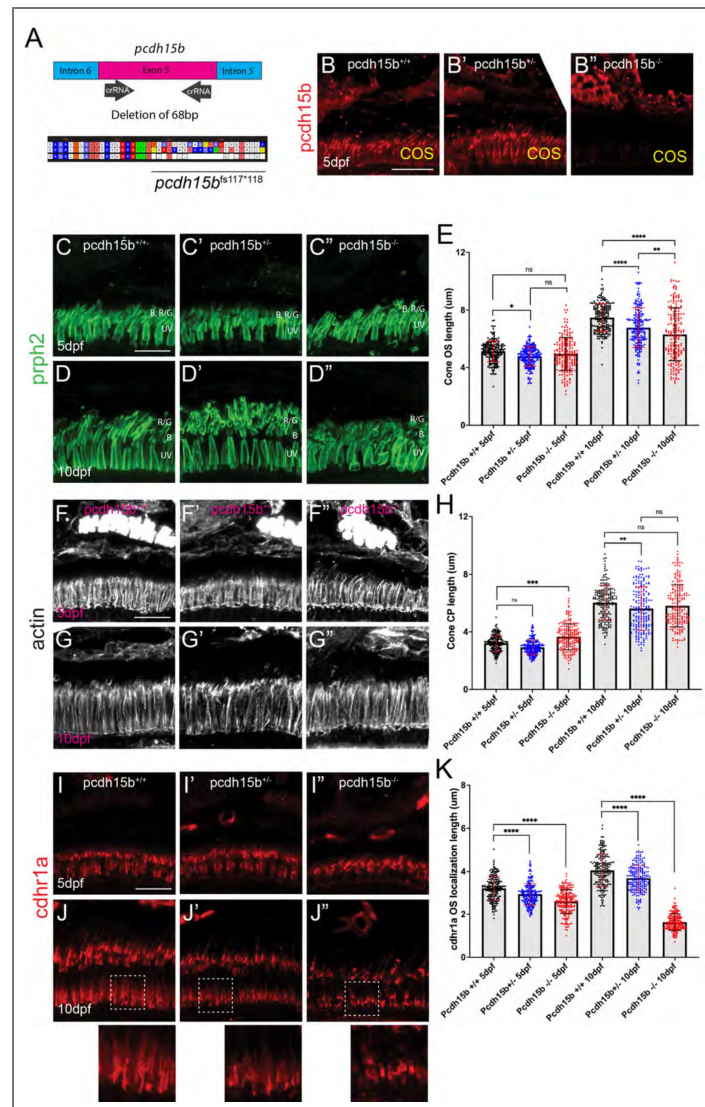


Figure 8. Loss of *pcdh15b* function leads to early cone OS defects and mis-localization of *cdhr1a*.

A) Diagrammatic representation of the CRISPR/Cas9 strategy for generating a *pcdh15b* loss of function allele. crRNAs were targeted to the flanking sequences of exon 5 and resulted in a heritable deletion of 68bp which resulted in a frameshift (fs) at AA117 and a premature stop codon (*) at AA118. **B-B'')** Confocal microscopy of retinal cryosections from wildtype, *pcdh15b*^{+/-} and *pcdh15b*^{-/-} individuals probed with *pcdh15b* antibody (red). Scale bar = 10µm. **C-D'')** Confocal microscopy of retinal cryosections from wildtype, *pcdh15b*^{+/-} and *pcdh15b*^{-/-} individuals probed with *prph2* antibody (green) to visualize the cone outer segments (COS) at 5dpf (C-C'') and 10dpf (D-D''). B=blue cones, RG=red/green cones, UV=UV cones. Scale bar = 10µm. **E)** Quantification of COS length using *prph2* signal for each genotype at each time point is depicted on the right. Standard deviation is shown in red. ns = not significant, ** = p<0.01. **** = p>0.0001. ANOVA = p<0.0001. **F-G'')** Confocal microscopy of retinal cryosections from wildtype, *pcdh15b*^{+/-} and *pcdh15b*^{-/-} individuals probed with actin antibody (white) to visualize cone CPs at 5dpf (F-F'') and 10dpf (G-G''). Scale bar = 10µm. **H)** Quantification of CP length using actin signal for each genotype at each time point is depicted on the right. Standard deviation is shown in red. ns = not significant, ** = p<0.01. **** = p>0.0001. ANOVA = p<0.0001. **I-J'')** Confocal microscopy of retinal cryosections from wildtype, *pcdh15b*^{+/-} and *pcdh15b*^{-/-} individuals probed with *cdhr1a* antibody (red) at 5dpf (I-I'') and 10dpf (J-J''). White boxes indicate regions enlarged. Scale bar = 10µm. **K)** Quantification of *cdhr1a* OS length for each genotype at each time point is depicted on the right. Standard deviation is shown in red. ns = not significant, *** = p<0.001. **** = p>0.0001. ANOVA = p<0.0001.

Loss of *cdhr1a* and *pcdh15b* synergize resulting in severe CP disruption and cone outer segment malformation

Having observed phenotypes affecting CPs and OSs in both *cdhr1a* and *pcdh15b* individual mutant lines, we next asked whether a combination of both mutations would lead to more severe outcomes. To do so, we established a *cdhr1a*^{-/-}; *pcdh15b*^{+/-} line. By crossing this line we were able to collect 5 and 10 dpf embryos that lacked both *cdhr1a* and *pcdh15b* (*cdhr1a*^{-/-}; *pcdh15b*^{-/-}) function. We then again performed our morphological analysis of cone OSs using prph2 antibody staining and CPs using actin antibody staining. At 5 dpf we observed little change in COS morphology when comparing *cdhr1a*^{-/-} to *pcdh15b*^{-/-} mutants (Fig 9A,B). COS length however differed, where *cdhr1a*^{-/-} individuals display significantly longer COSs at 5.63μm versus 4.96 for *pcdh15b*^{-/-}. Interestingly, when comparing the double mutant to either single mutant, or any of the other genetic combinations we observed a significant decrease in COS length, 3.68μm, as well as disorganized COS morphology (Fig 9A-D). This trend was even more pronounced at the 10 dpf timepoint, where disorganization of the COS was highly evident in the double mutant (Fig 9E-H). Furthermore, the double mutant COS length decreased to 4.31μm compared to 7.83μm for *cdhr1a*^{-/-} and 6.31μm for *pcdh15b*^{-/-} (Fig 9 E-I). Thus, it appears that concurrent loss of *cdhr1a* and *pcdh15b* combines to result in significant decreases in COS length as well as very early morphological COS disorganization.

When examining CP length and morphology we observed similar trends. At 5 dpf, *cdhr1a* mutants exhibit elongated CPs with normal morphology, compared to slightly truncated but morphologically normal CPs in *pcdh15b* mutants (Fig 9K,L). Double mutants display normal CP morphology at 5 dpf, albeit their CPs are significantly shorter than wildtype (3.22μm vs 2.45 μm) (Fig 9N, S). By 10 dpf both CP morphology and length are significantly affected in the double mutant compared to either single mutant (Fig 9O-P, T). CPs appear severely disorganized lacking the expected vertical actin staining, instead appearing shorter and bent or tilted. When comparing CP lengths, the double mutant has severely truncated CPs, 2.93μm vs 6.34μm in *cdhr1a* mutants and 5.82μm in *pcdh15b* mutants. Taken together, we conclude that the combination of *pcdh15b* and *cdhr1a* loss of function enhances both COS disorganization and decreases COS length along with CP morphological disruption and severe truncation even at very early timepoints. These findings further support our hypothesis where *cdhr1a* functions to link cone OSs with CPs by interacting with *pcdh15b*.

Discussion

Discovered over 20 years ago, the retinal cadherin CDHR1 is clinically associated with recessive cone-rod dystrophy, yet its exact molecular function continues to remain a mystery. In our current study we found that the zebrafish homologue of CDHR1, *cdhr1a*, localizes to cone and rod OSs in a pattern mirroring that of the inner segment calyceal processes and its own resident cadherin *pcdh15b* (PCDH15 homolog in zebrafish) (Figure 1). Interestingly, the pattern of CDHR1/PCDH15 localization was observed across various species, including amphibians, rodents, birds, primates and humans, thus indicating a high degree of mechanistic evolutionary conservation (Figure 2). Aquatic species, zebrafish and xenopus, appear to have retained significant length of their cone CPs, particularly along the UV cones, as evident by the length of *cdhr1/pcdh15* signal (Fig 1C, E). Conversely in rodents, which have been previously shown to either lack CPs or contain a single vestigial structure, CDHR1/PCDH15 juxtaposition is still maintained albeit in a very short segment (23). Intriguingly, the size of the CP may be related to the diurnal nature of an organism as the gerbil exhibited longer structures compared to rats and mice (Fig 2H). Birds and primates exhibit CP lengths longer than rodents but still shorter than fish or frogs (Fig 1K-L). In addition to localization, we showed that *cdhr1a* physically interacts with *pcdh15b* *in vitro*, via co-immunoprecipitation and *in vivo* via the K562 cell aggregation assay (Fig 3). Furthermore, using immuno-gold TEM we observed direct interaction between *cdhr1a* and *pcdh15b* at the photoreceptor-CP interphase region. Taken together, the localization and interaction findings provide the first potential clue to the molecular function of CDHR1 and the etiology of CDHR1-

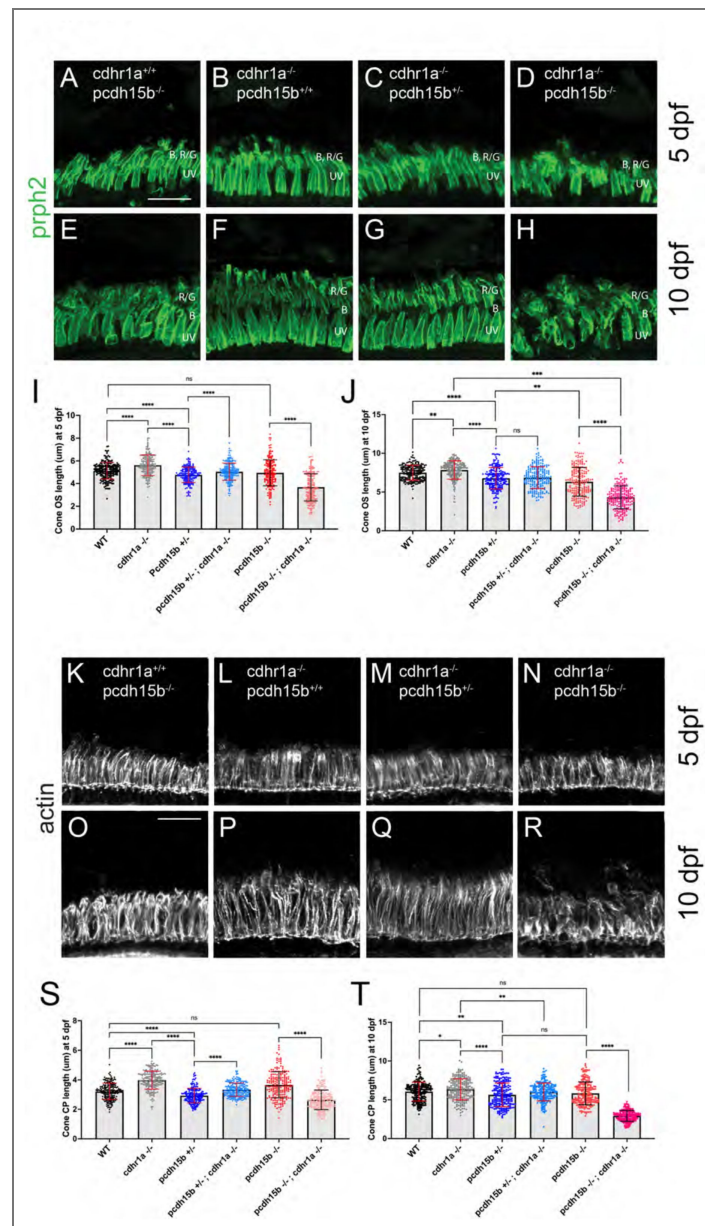


Figure 9. Cone phenotypes are exacerbated by simultaneous loss of pcdh15b and cdhr1a.

A-D) Confocal microscopy of retinal cryosections from 5 dpf *pcdh15b*^{-/-}, *cdhr1a*^{-/-}, *cdhr1a*^{-/-}; *pcdh15b*^{+/-} and *cdhr1a*^{-/-}; *pcdh15b*^{-/-} larva probed with prph2 antibodies (green). COS = cone outer segment. B=blue cones, RG=red/green cones, UV=UV cones. Scale bar = 10µm. **E-H)** Confocal microscopy of retinal cryosections from 10 dpf *pcdh15b*^{-/-}, *cdhr1a*^{-/-}, *cdhr1a*^{-/-}; *pcdh15b*^{+/-} and *cdhr1a*^{-/-}; *pcdh15b*^{-/-} larva probed with prph2 antibodies (green). COS = cone outer segment. B=blue cones, RG=red/green cones, UV=UV cones. Scale bar = 10µm. **I)** Quantification of cone outer segment length at 5 dpf based on prph2 signal. ns = not significant, **** = p<0.0001. ANOVA = p<0.0001. **J)** Quantification of cone outer segment length at 10 dpf based on prph2 signal. ns = not significant, ** = p<0.01 ****, =p<0.0001. ANOVA = p<0.0001. **K-N)** Confocal microscopy of retinal cryosections from 5 dpf *pcdh15b*^{-/-}, *cdhr1a*^{-/-}, *cdhr1a*^{-/-}; *pcdh15b*^{+/-} and *cdhr1a*^{-/-}; *pcdh15b*^{-/-} larva probed with actin antibodies (white). COS = cone outer segment. Scale bar = 10µm. **O-R)** Confocal microscopy of retinal cryosections from 10 dpf *pcdh15b*^{-/-}, *cdhr1a*^{-/-}, *cdhr1a*^{-/-}; *pcdh15b*^{+/-} and *cdhr1a*^{-/-}; *pcdh15b*^{-/-} larva probed with actin antibodies (white). COS = cone outer segment. Scale bar = 10µm. **S)** Quantification of cone CP length at 5 dpf based on prph2 signal. ns = not significant, **** = p<0.0001. ANOVA = p<0.0001. **T)** Quantification of cone CP length at 10 dpf based on prph2 signal. ns = not significant, * = p<0.05 ** = p<0.01 ****, =p<0.0001. ANOVA = p<0.0001.

associated cone-rod dystrophy. Based on these findings we therefore hypothesized that *cdhr1a* links the OSs with the CPs via interactions with *pcdh15b* and postulate the absence of this interaction as a mechanism driving cone cell degeneration.

To test our hypothesis, we tracked cone-rod dystrophy progression in zebrafish using single *cdhr1a*^{-/-} or *pcdh15b*^{-/-} mutant lines as well as a *cdhr1a*^{-/-}, *pcdh15b*^{-/-} double mutants. Taking advantage of the zebrafish system we were able to examine CRD progression at numerous developmental and adult stages at the molecular level with high resolution. When examining the *cdhr1a*^{-/-} model, we documented progressive disorganization and shortening of cone OS starting at 30 dpf, while observing only minor effects in rods (Fig 5 [E](#), 6 [E](#)). Interestingly, disorganization and shortening of cone OSs at 30 dpf was preceded by slight but significant lengthening at 5 and 10 dpf. This suggests that early development of the OS may be regulated by CP-OS interactions to ensure proper rate of OS growth (Fig 5H [E](#)). *Cdhr1a*^{-/-} cones exhibited shortened and disorganized CPs as early as 5 dpf, indicating that defects in CP-OS interactions precede and may therefore lead to cone OS instability (Fig 7 [E](#)). However, we also observed mutant CP overgrowth at 15dpf suggesting that CP assembly or maintenance is mis-regulated in the absence of *cdhr1a* (Fig 7O [E](#)). A striking phenotype observed in mutant cone OSs was the mis organization of *prph2* signal. Normally *prph2* signal appears smoothly along the periphery of cone OS and largely absent from the center, however, in *cdhr1a*^{-/-} cones we observe an almost disc like horizontal staining of *prph2* throughout the OS as early as 15 dpf and very clearly by 30dpf (Fig 5B [E](#)-G'). *Prph2* mis organization was also accompanied by the appearance of cone OS bending and tilting, compared to wildtype (Fig 5A-G [E](#)). Similar *prph2* localization phenotypes have been observed in *prom1b* mutant zebrafish, which is interesting as *prom1* has been shown to interact with *Cdhr1* [15](#). Perhaps loss of *prom1b* results in the mis-localization or mis-regulation of *cdhr1a* in the cone OS and therefore also compromises CP-OS interactions leading to similar phenotypes. Finally, using TEM, we observed that *cdhr1a*^{-/-} cone OSs displayed disorganized disc stacking, which supports the observed *prph2* localization phenotype (Fig 5K,L [E](#)). Compared to previous studies of *cdhr1* function in frogs, or *pcdh15b* function in zebrafish, we did not observe disc overgrowth in 15dpf zebrafish. This may result from disorganization and loss of OS integrity already by 15dpf. Disc overgrowth may therefore be observed at earlier timepoint but those were not examined in our studies. Several previous studies have postulated that CP-OS interactions are critical for stability of cone OSs [23,24,32](#). Our data supports this notion where *cdhr1a* mutant cone OSs are shorter and often appear tilted to one side. Similar OS phenotype results were also observed in *pcdh15b* mutants up to 10dpf which is in line with previous reports from zebrafish and what was also observed in *xenopus* embryos injected with *Pcdh15* morpholinos [31,32](#). Surprisingly *pcdh15b* heterozygotes exhibited slight but significant reduction in cone OS and CP length (Fig 8E, H [E](#)). This suggests that steady state levels of *pcdh15b* protein are critical for its function. *Pcdh15b* mutants display cone OS defects earlier than *cdhr1a*^{-/-} and do not exhibit the early increase in OS length (Fig 8 [E](#)). This might suggest that the absence of *pcdh15b* interferes with CP function, while the absence of *cdhr1a* interferes with OS-CP connections but not the CP itself. In humans *PCDH15* mutations are associated rod-cone dystrophy, however, due to lethality of the *pcdh15b* mutation by 15 dpf, we are not able to compare long-term *pcdh15b* mutant phenotypes with *cdhr1a* mutants. In the double mutants, due to absence of both cadherins, we observed shorter and disorganized CPs and cone OSs as early as 5 dpf, normally not seen in either single mutant alone (Fig 9 [E](#)). By 10 dpf the double mutant phenotypes were severe and comparable to what we observed in *cdhr1a*^{-/-} mutants by 30+ dpf (Fig 9 [E](#)). The exacerbation of phenotypes in the double mutant further supports our model where *cdhr1a* and *pcdh15b* function to anchor cone OSs with CPs. Interestingly, these results also suggest that in the absence of either *cdhr1a* or *pcdh15b* there may be redundant mechanisms that aid in OS-CP anchoring. Perhaps other retinal cadherins can interact with *cdhr1a* or *pcdh15b*, albeit not as strongly or efficiently, and delay the early phenotypes until the OSs are fully mature. Taken together, our data along with previous studies strongly supports the hypothesis that in the absence of *cdhr1a* function the CP-OS interaction is compromised and thus hampers cone OS integrity.

While the focus of our study was on cone cells, we also characterized effects of *cdhr1a* loss on rods. Rod OSs, nor their CPs appear to be significantly affected by the loss of *cdhr1a* function until later stages, 3 months or longer (Fig 6, 7). Albeit significant, these effects were much less obvious compared to the effects on cones, as one might expect for a cone-rod dystrophy model. Furthermore, while cone CPs displayed gross morphological defects, rod CPs were only slightly affected even at later timepoints (Fig 6, 7). This likely stems from the fact that rod CPs are inherently much smaller than cone CPs and may not serve as critical a role in rods as we predict they do in cones^{23,24,31,32}. Comparing rod CP length over the course of development and adulthood in our study we note that rod CPs are on average approximately one third the size of cone CPs (Fig 7). Furthermore, compared to the length of rod OS, rod CPs encompass a very small portion of the OS, while cone CPs extend up to three quarters of the cone OS length.

Based on these observations of rod CPs, we hypothesize that zebrafish rod CPs only extend along the newly forming OS discs and do not provide structural support to the ROS. As such, we support the hypothesis postulated by Burgoyne 2015 *et al* that in rods, *cdhr1a* functions to anchor newly formed OS discs to facilitate proper maturation of the discs prior to release into the OS. While this differs from its function in cones, *cdhr1a* still likely relies on its interaction with *pcdh15b* to serve this function as the two proteins clearly juxtapose in rod cells across all the species we examined, including rodents which have been hypothesized to lack CPs or have very simplified versions of CPs compared to other species²³.

Discovery of the interaction between *cdhr1a* and *pcdh15b* is novel. Previous studies have suggested that PCDH15 and CDH23 are binding partners in the retina as they are well known to interact in the inner ear stereo cilia³². However, our work, and others, have shown that *cdh23* is not expressed in zebrafish retina nor does loss of *cdh23* result in ocular phenotypes (Fig S1).

Work from macaque retina shows *cdh23* expression in the OS/IS region but does not display the prototypical CP staining as observed with *pcdh15* or actin²³. While several Usher 1 proteins other than *pcdh15* have been found to localize to the CP, including harmonin, myosin and sans, there has been a missing link between *pcdh15* in CP and its potential partner the OS, which we now propose to be *Cdhr1*. In our study, *cdhr1a*-*pcdh15b* interactions are supported by immunoprecipitation and cell aggregation assays and by the fact both are cadherin proteins that can facilitate heterophilic interactions (Fig 3). PCDH15 is known to generate cis dimers like *cdh23*, and as such the *cdhr1a*-*pcdh15b* interaction may in fact involve hetero-tetrameric complexes³⁶. Additional evidence for the *cdhr1a*-*pcdh15b* interaction was the observation that *cdhr1a* localization along the cone OS was significantly impaired in the absence of *pcdh15b* function. At 5 dpf and even more so at 10dpf the length of *cdhr1a* signal along the cone OS was severely reduced, while cone OS length was only slightly affected (Fig 9). Thus, we conclude that while *pcdh15b* is not necessary for *cdhr1a* to localize to the base of cone OSs, its absence prevents *cdhr1a* from forming connections to the CP and subsequently extending along the length of the OS. Other proteins may further regulate *cdhr1a*-*pcdh15b* complexes such as *prom1b*, which is known to interact with *cdhr1*, as well as other Usher 1 proteins found in the CP. CDHR1 is also known to undergo proteolytic cleavage to remove its extracellular cadherin domains, which could further regulate the interaction and stability of the CP-OS attachments⁹. It remains unclear if the cleavage occurs in both rod and cone cells. We hypothesize that proteolytic cleavage is specific to rods, where this would aid in regulating the release of the newly matured OS discs, while cleavage would be absent in cone cells as to ensure CP-mediated OS structure and integrity. Interestingly, we have also previously reported that *cdhr1a* can be targeted by the *Siah1* ubiquitin ligase for proteasomal degradation, adding yet another layer to the potential complexity of regulating *cdhr1a*³⁷. Imaging of *siah1* indicates that it also localizes to the OS in a pattern similar to that of *cdhr1a* (data not shown). Future studies will be required to elucidate the entire complex responsible for the anchoring and maintaining of CPs and OSs in rods versus cones.

Overall, our *cdhr1a*^{-/-} model corroborates findings from the mouse CDHR1 KO model with early phenotypes found exclusively in cone cells, and only at later stages, 6+ months, displaying rod cell OS disorganization and degeneration. In conclusion, we propose that *cdhr1a* and *pcdh15b* function to link the cone OS and the CP to maintain cone homeostasis in an evolutionarily

conserved manner. Disruption of this interaction may therefore be a potential driving factor in the progression of cone-rod dystrophy observed in CDHR1 patients. Future studies will need to focus on the molecular consequences of OS-CP interaction and the identity of the entire repertoire proteins involved as well as any potential signaling pathways that activate in response to sensing or monitoring OS-CP interactions.

Materials & methods

Zebrafish Husbandry and Embryo maintenance

Zebrafish husbandry used in all procedures were approved by the University of Kentucky Biosafety office as well as IACUC Policies, Procedures, and Guidelines and ethical standards. The AB strain was used as wildtype. ALT-R-CRISPR Cas9-based mutant lines were generated based on protocols established previously³⁵. Embryos were kept at 28°C in E3 embryo media. At indicated times in the study, embryos and adults were euthanized by cold shock or in tricaine before harvesting the eyes and fixation with 4% PFA in PBS overnight at 4°C.

Statistical analysis

All the data sets were analyzed using Prism 10. Data shown on graphs represents individual measurements $\pm$ standard deviation. Each of the data points had an n of 5+ individuals. One-way ANOVA with Tuckey post hoc multiple comparisons analysis was used for assessing direct comparison significance between ages or genotypes. All of the graphs depict the mean value.

Whole mount In Situ Hybridization

WISH was performed as previously described³⁸. RNA probes were generated via PCR amplification from 3 dpf cDNA fused to T7 promoter sequence and subsequently transcribed (DIG or FITC labeled) using T7 polymerase (Roche). Primer sequences can be found in [supplemental table 1](#)³⁸. Approximately 1000bp was amplified from the 3' end of each cDNA. WISH was performed as previously described³⁸. Images were captured using a Nikon Digital sight DS-U3 camera and Elements software. Image adjustment was performed using Adobe Photoshop.

Cryosection and Immunofluorescence/IHC

Embryos and adult eyes were fixed in 4% paraformaldehyde then washed out PBS with 0.1% Tween-20. Next, the specimens were washed overnight in 10% then 30% sucrose overnight at 4°C. Transverse, 10 mm sections were collected, beginning just anterior to and ending posterior to the eye. For imaging and cell quantification, only the sections containing the lens were used for consistency. Immunofluorescence was performed as described previously with addition of a Tris-EDTA based antigen retrieval step prior to serum blocking the specimen³⁵. Post blocking, following antibodies and lectins were used: anti-zcdhr1a (Cdhr1a, rabbit, 1:100, SKU:DZ07988 Boster Bio, Pleasanton, CA, United States), anti-hcdhr1 (CDHR1, rabbit, 1:100, PA5-57832 Thermo Fisher, Waltham, MA, United States), anti-hpcdh15 (Pcdh15, sheep, 1:75, PA5-47865 Thermo Fisher, Waltham, MA, United States), anti-prph2 (Peripherin – 2, rabbit, 1:100, 18109-1-AP Protein-tech, Rosemont, IL, United States), anti-actin (beta Actin, mouse, 1:100, MA1-140 Thermo Fisher, Waltham, MA, United States) anti-gnb1 (GNB1, rabbit, 1:100, PA5-30046 Thermo Fisher, Waltham, MA, United States), anti-zrhodopsin (1d1 epitope of the zebrafish rhodopsin, mouse 1:100, Fadool lab, Florida State University, FL, United States). Alexa fluor 488, 555, and 647 conjugated secondary antibodies at a concentration of 1:200 (Thermo Fisher, Waltham, MA, United States) were used with DAPI as a counter stain. Further, Lectins: Peanut Germ Agglutinin conjugated with a 488 fluorophore (1:100 concentration, #29060 Biotium, San Francisco, CA, United States) and Wheat Germ Agglutinin conjugated with a 405 fluorophore (1:50 concentration, #29028-1 Biotium, San Francisco, CA, United States) were used to label PRC outer segments.

Confocal/SIM imaging

For Immunofluorescence, slides were mounted in Vectashield antifade mounting medium (H-1700-10 Vector laboratories, Newark, CA, United States) then imaged on a Nikon C2 confocal microscope under a 60X 1.4 NA oil immersion objective. For consistency all the images were captured from the central retina to assess all the major types of photoreceptors. Super resolution microscopy was performed on a Nikon A1R confocal microscope equipped with a Structured Illumination Microscopy module and a CMOS sensor under a 100X 1.49 NA oil immersion objective. Quantification of OS length and number was performed manually using confocal captured images in Image J.

Cloning

A previously developed and verified *Cdhr1a* mammalian cell expression plasmid of *Cdhr1a* tagged with FLAG in pCIG2 developed via Infusion HD cloning plus (Takara) was used in this study³⁵. To develop an expression construct for *Pcdh15b*, we amplified *pcdh15b* cDNA from 3dpf WT embryos (5'CATCATGATATCACCATGAAGATGCGCCAGAGGTCG-3', 5'ATGATGTTCTGAACAGAACAGTGGACTGAGATGG3') and cloned into the Topo XL vector (Invitrogen). Subsequently we directionally cloned *pcdh15b* cDNA into the pCDNA3-MYC expression plasmid using Kpn1 and Xho1. Both constructs were verified using sanger sequencing (Eurofinsgenomics).

Immunoprecipitation

For transient mammalian cell transfection a previously established protocol was used with slight modifications³⁹. First, HEK 293 cells were cultured at 37 °C in DMEM media until 70% confluency. Next, aforementioned plasmids individually or in combinations of *Cdhr1a*:FLAG pCIG2 (1ug) and *Pcdh15b*:MYC pCDNA (1ug) were first incubated with PEI (polyethyleneimine) at room temperature for 30 mins. Upon HEK 293 cell confluency, these PEI/DNA complexes were transfected to the cells and incubated at 37°C for 24 hours. Protein extraction was performed based on the Mem-PER plus extraction protocol (#89842 Thermo Fisher). To inhibit protease activity, we added HALT protease inhibitor (1:1000 #78425 Thermo Fisher). For co-immunoprecipitation (co-IP), one *Cdhr1a* and *Pcdh15b* cotransfected protein sample was incubated with Anti-FLAG magnetic beads (A36797 Thermo Fisher) while another cotransfected sample was incubated with Anti-MYC magnetic beads (#88842 Thermo Fisher). For IP control, *Cdhr1a*-FLAG transfection samples were immunoprecipitated (IP) using Anti-MYC beads while *Pcdh15b*-MYC transfection samples were immunoprecipitated using Anti-FLAG beads. Next, all the lysate control, control IP, and co-IP samples were prepared with 6X SDS sample buffer under reducing conditions and boiled at 96°C for 7 mins. Samples were then ran in two 7.5% SDS PAGE gels at 140 volts for 90 mins. They were then transferred to a PVDF membrane for 90 mins at 20 volts. The membranes was then blocked with 1X BlockerTM (#37565 Thermo Fisher) for 1 hour at room temperature. Next, one membrane was probed for FLAG via mouse Anti-FLAG (1:1000 F1804 Sigma Aldrich, St. Louis, MO, United States) and the other membrane was probed for MYC via mouse Anti-MYC (1:1000 MA121316 Thermo Fisher). These primary antibodies were incubated after blocking at 4°C overnight. The following day membranes were washed 3 times with TBS (5 mins each). Subsequently, the membranes were incubated with goat anti-mouse secondary antibody conjugated with poly HRP at 1:2000 concentration in the 1X BlockerTM for 1 hour at room temperature. Next, the membranes were again washed with TBS buffer for 3 times (5 mins each). To detect the HRP signal, SuperSignalTM West Pico PLUS Chemiluminescent Substrate (#34580 Thermo Fisher) was used and chemiluminescence images were then detected and captured using Amersham Imager 680.

K562 assay

For the cell aggregation assay, K562 cells were cultured in plastic dishes at 37°C in RPMI 1640 media until 70% confluency. *Cdhr1a* and *Pcdh15b* expression plasmids were incubated with the PLUS reagent for 15 mins in the same RPMI 1640 media and then Lipofectamine LTX was added and incubated for 25 mins at room temperature to form Lipofectamine-DNA complexes. The

complexes were then transferred to the cells and incubated at 37°C for 24 hours. Next day individual transfections of *Cdhr1a*, *Pcdh15b* or control were co-cultured by isolating each population of cells from the dishes, mixing them together and re-plating to allowing aggregates to form. Any K562 cell aggregates that formed were then imaged 24h later using the EVOS FLoid cell culture microscope (#4471136 Thermo Fisher). The experiment was repeated three times and images from five different fields of view of each replicate were quantified.

TEM

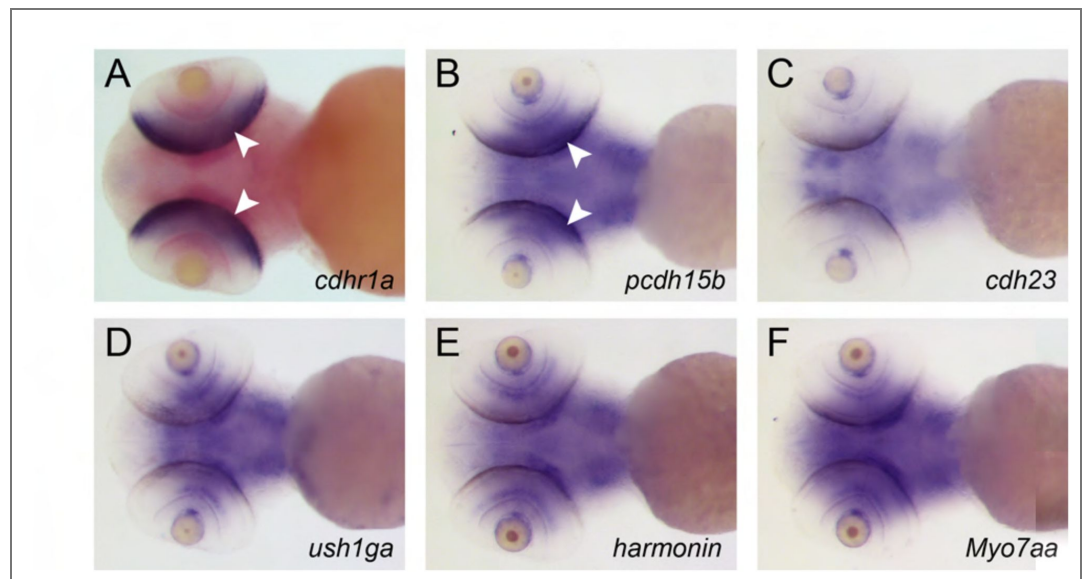
TEM sample preparation was slightly modified from *Miles et al.* 2021. First, zebrafish larvae and adult specimens were fixed in 2.5% glutaraldehyde and 2% paraformaldehyde in 0.1M phosphate buffer at 4°C overnight on a shaker. Samples were then washed 5 times with 0.1M phosphate buffer (with 0.1% Tween-20) for 5 times - 10 minutes each to wash out the aldehyde fixatives. Next, samples were placed in 1% Osmium tetroxide solution for 1.5 hours in dark at room temperature on a shaker. Samples were then washed again with 0.1M phosphate buffer with Tween-20 for 5 times - 5 minutes each and subsequently dehydrated with a series of ethanol washes at 50% (10mins), 70% (10 mins), 80% (10 mins), 90% (15 mins), and two 100% ethanol washes (15 mins each). Next, the specimens were infiltrated with LR white resin through a series of ethanol:resin combinations 25% LR white:75% ethanol (30 mins), 50% LR white:50% ethanol (30 mins), 75% LR white:25% ethanol (1 hour), and with 100% LR white (overnight at 4°C). The following day samples were exchanged with fresh 100% LR white for 3 hours and then samples were placed in "00" gelatin capsules or sealed PCR tubes in order to perform anaerobic polymerization required for LR white at 65°C. One micron thin sections were collected via a glass blade from the resin blocks using Leica EM UC7 and were stained with Toluidine blue to assess the location of samples being sectioned. Once satisfied with the location, ultrathin 70nm sections were collected using a diamond knife (Ultra 45° DiATOME, Quakertown, PA, United States) and placed on slotted EM grids (G2010-Ni Electron Microscopy sciences, Hatfield, PA, United States) covered in a layer of 0.5% formvar. Sections were then stained with a non-radioactive heavy metal UranylLess EM stain (#22409 Electron Microscopy sciences) for 1 hour followed by Reynold's lead citrate stain for 30 mins. Images were then collected using FEI Talos F200X at 200 keV.

Immuno-TEM

5dpf embryos were fixed in 4% paraformaldehyde then washed-out using PBS with 0.1% Tween-20. Next, eyes were enucleated from embryos and were processed for whole mount Immunohistochemistry (IHC). Whole mount IHC was performed as described previously, but with exception where heat mediated antigen retrieval for whole eyes was conducted in a thermal cycler prior to serum blocking. Post blocking, primary antibodies used were: anti-zdhr1a (SKU: DZ07988 Boster Bio, Pleasanton, CA, United States), and anti-hpcdh15 (PA5-47865 Thermo Fisher, Waltham, MA, United States). Gold conjugated secondary antibodies used were: goat anti-Rabbit Alexa Fluor 488-5 nm colloidal gold (A-31565, Thermo Fisher, Waltham, MA, United States) and Donkey-Anti-Sheep 25 nm colloidal gold (SKU: 25829 Electron Microscopy sciences, Hatfield, PA, United States). Gold labeled eyes were then processed for TEM as described previously.

CRISPR

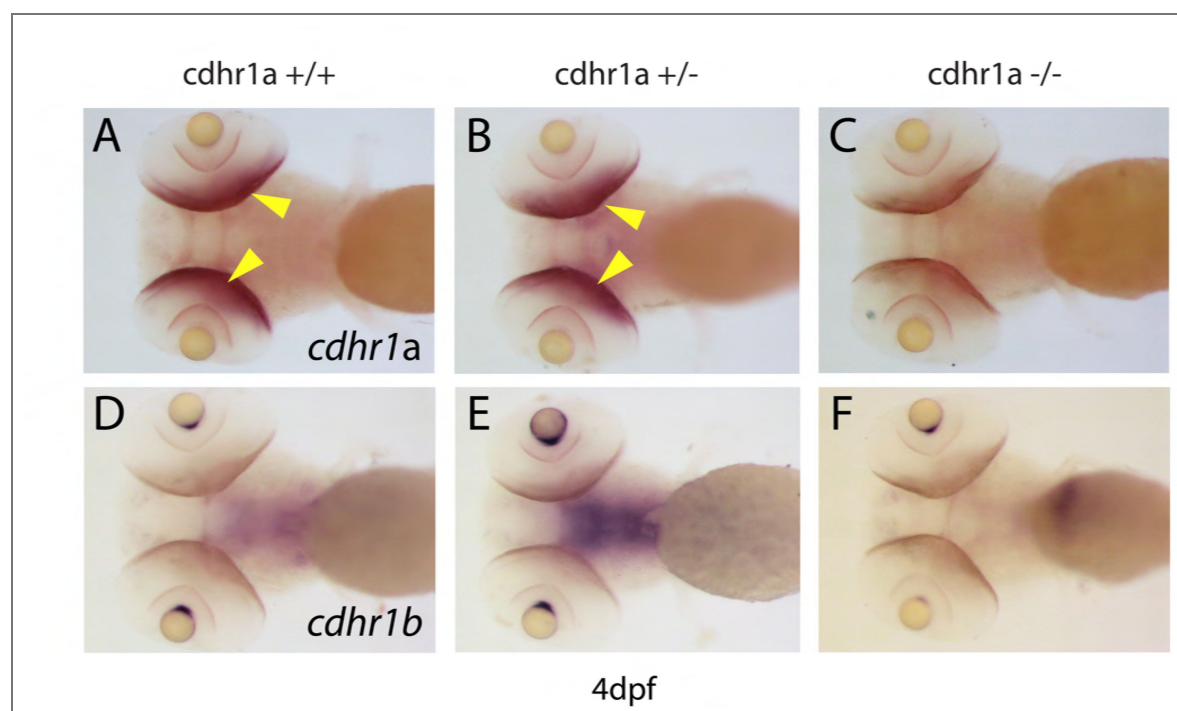
To establish the CRISPR/Cas9 based mutant lines, we ordered predesigned and synthesized ALT-R Crispr crRNA (IDTDNA) targeting *Cdhr1a* (GTCTGGAAGTAGCATCTATA, TCTGGCACATCTACGATGGA) and *Pcdh15b* (CACCACAATGGACTGGATGT,CGACTATCCGCACCTCGTGT). Next crRNA-tracrRNA duplexes were constructed before injecting with Alt-R Cas9 v.3 enzyme in single cell staged embryos as described previously⁴⁰. Genotyping was performed by designing primer oligos upstream and downstream of the expected CRISPR/Cas9 based deletion sites and the resulting difference in sequence size was used to genotype using DNA gel electrophoresis. (*Cdhr1a*: 5'-GTGTT AAAATTTGAATGCTGAG-3', 5'-CTGCATATGCTTAGATGTTACC-3', *pcdh15b*: 5'-GAAACACAAAAGAAGCT GCG-3', 5'-GCCTTTATAATGGAGCGCAAG-3') Both *Cdhr1a* and *Pcdh15b* deletion sequences were then confirmed via Sanger sequencing after outcrossing for two filial generations.



Supplementary figure 1. Expression of USHR genes in the zebrafish retina. Whole mount in situ hybridization of 5 dpf zebrafish larva for **A)** *cdhr1a*, **B)** *pcdh15b*, **C)** *cdh23*, **D)** *ush1ga*, **E)** *harmonin*, **F)** *myo7aa*. Ventral images of the embryos are displayed. White arrows indicate expression in the outer nuclear layer of the retina.

Supplementary figure 2. Expression of *cdhr1a* and *cdhr1b*.

Whole mount in situ hybridization of 5 dpf zebrafish larva for **A-C)** *cdhr1a*, **D-F)** *cdhr1b* in WT, *cdhr1a*^{+/-} or *cdhr1a*^{-/-} individuals. Ventral images of the embryos are displayed. Yellow arrows indicate expression in the outer nuclear layer of the retina. *cdhr1a* retinal expression is lost in the homozygous mutant. *cdhr1b* expression is absent from the retina in wildtype and *cdhr1a* mutants.



Supplementary Table 1. WISH probe primer sets

<i>cdhr1a</i>	ATGAAGAATGCAAGGGAAATA	TAATACGACTCACTATAGGGTCCTTCTGGACTGATTCCAATGC
<i>pcdh15b</i>	GGTGATGGATCCAGTTCAGTG	TAATACGACTCACTATAGGGTCACAGAACAGTGGACTGAGA
<i>pcdh23</i>	AGTGATTCAGATGATCGACGA	TAATACGACTCACTATAGGGTCATAACTCTGTGATCTCTAA
<i>harmonin</i>	CCTTAGTTTGGTGGGCACCAA	TAATACGACTCACTATAGGGTTAAAAGAATGTCACCTCATC
<i>ush1ga</i>	TTCTTGCCTTAATGTCTGTTT	TAATACGACTCACTATAGGGGCGCAGCTTTCACAAAACCAT
<i>myo7aa</i>	AAACAAGGACATTTAACCAC	TAATACGACTCACTATAGGGGAGTCACATCGATCACTGGAC

Data availability

All data generated or analyzed during this study are included in this published article. All materials, zebrafish lines, and reagents will be shared upon request after publication.

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Additional information

Author contributions

JKF and MP designed the experiments. JKF and MKP wrote and edited the manuscript. Experiments were carried out by MKP and WP.

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KYINBRE core facility voucher	P20GM103436	Jakub K Famulski

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Peer reviews

Reviewer #1 (Public review):

Mutations in CDHR1, the human gene encoding an atypical cadherin-related protein expressed in photoreceptors, are thought to cause cone-rod dystrophy (CRD). However, the pathogenesis leading to this disease is unknown. Previous work has led to the hypothesis that CDHR1 is part of a cadherin-based junction that facilitates the development of new membranous discs at the base of the photoreceptor outer segments, without which photoreceptors malfunction and ultimately degenerate. CDHR1 is hypothesized to bind to a transmembrane partner to accomplish this function, but the putative partner protein has yet to be identified.

The manuscript by Patel et al. makes an important contribution toward improving our understanding of the cellular and molecular basis of CDHR1 associated CRD. Using gene editing, they generate a loss of function mutation in the zebrafish *cdhr1a* gene, an ortholog of human CDHR1, and show that this novel mutant model has a retinal dystrophy phenotype, specifically related to defective growth and organization of photoreceptor outer segments (OS) and calyceal processes (CP). This phenotype seems to be progressive with age. Importantly, Patel et al, present intriguing evidence that *pcdh15b*, also known for causing retinal dystrophy in previous *Xenopus* and zebrafish loss of function studies, is the putative *cdhr1a* partner protein mediating the function of the junctional complex that regulates photoreceptor OS growth and stability.

This research is significant in that it:

- (1) provides evidence for a progressive, dystrophic photoreceptor phenotype in the *cdhr1a* mutant and, therefore, effectively models human CRD; and
- (2) identifies *pcdh15b* as the putative, and long sought after, binding partner for *cdhr1a*, further supporting the theory of a cadherin-based junction complex that facilitates OS disc biogenesis.

Comments on the revised version of the manuscript:

The authors adequately addressed previous comments related to lack of details on quantitative and statistical analyses and methods. In this regard, I believe the revised manuscript presents a stronger analysis of the data. I also appreciated the revised discussion section, which better contextualizes their new data with previous observations in different animal models.

The authors provided additional evidence in Fig 1C-H for the co-localization of *pcdh15b* and actin within CPs using immunolabeling with super resolution imaging. This data firmly supports their other observations. A similar approach tends to also show co-localization of actin and *cdhr1a*, although the authors suggest that the pattern of expression is less overlapping, which would be expected if *cdhr1a* is predominately expressed in the OS membranes whereas *pcdh15b* is predominantly expressed in the CP membranes. In my opinion the data presented to support this separation is still not that convincing. Moreover, the authors show that both *cdhr1a* and *pcdh15b* are expressed in CPs using immuno-TEM (Fig 1I). This is a difficult question to address experimentally, and it is, of course, still plausible that *pcdh15b* within the CP membrane and *cdhr1a* within the OS membrane are interacting in trans. However, I just don't think that the data unequivocally support mutually exclusive localization of these proteins as suggested by the authors and depicted in the model in Fig 1J.

<https://doi.org/10.7554/eLife.102258.2.sa3>

Reviewer #2 (Public review):

Summary:

The goal of this study was to develop a model for CDHR1-based Con-rod dystrophy and study the role of this cadherin in cone photoreceptors. Using genetic manipulation, a cell binding assay, and high- resolution microscopy the authors find that like rods, cones localize CDHR1 to the lateral edge of outer segment (OS) discs and closely opposes PCDH15b which is known to localize to calyceal processes (CPs). Ectopic expression of CDHR1 and PCDH15b in K652 cells indicate these cadherins promote cell aggregation as heterophilic interactants, but not through homophilic binding. This data suggests a model where CDHR1 and PCDH15b link OS and CPs and potential stabilize cone photoreceptor structure. Mutation analysis of each cadherin results in cone structural defects at late larval stages. While *pcdh15b* homozygous

mutants are lethal, *cdhr1* mutants are viable and subsequently show photoreceptor degeneration by 3-6 months.

Strengths:

A major strength of this research is the development of an animal model to study the cone specific phenotypes associate with CDHR1-based CRD. The data supporting CDHR1 (OS) and PCDH15 (CP) binding is also a strength, although this interaction could be better characterized in future studies. The quality of the high-resolution imaging (at the light and EM levels) is outstanding. In general, results support the conclusions of the authors.

Weaknesses:

While the cellular phenotyping is strong, the functional consequences of CDHR1 disruption is not addressed. While this is not the focus of the investigation, such analysis would raise the impact of the study overall. This is particularly important given some of the small changes observed in OS and CP structure. While statistically significant, are the subtle changes biologically significant? Examples include cone OS length (Fig 4F, 6E) as well as other morphometric data (Fig 7I in particular). Related, for quantitative data and analysis throughout the manuscript, more information regarding the number of fish/eyes analyzed as well as cells per sample would provide confidence in the rigor. The authors should also not whether analysis was done in an automated and/or masked manner.

Comments on revisions:

Most of my concerns were addressed in this revised version.

<https://doi.org/10.7554/eLife.102258.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Patel et al investigates the hypothesis that CDHR1a on photoreceptor outer segments is the binding partner for PCDH15 on the calyceal processes, and the absence of either adhesion molecule results in separation between the two structures, eventually leading to degeneration. PCDH15 mutations cause Usher syndrome, a disease of combined hearing and vision loss. In the ear, PCDH15 binds CDH23 to form tip links between stereocilia. The vision loss is less understood. Previous work suggested PCDH15 is localized to the calyceal processes, but the expression of CDH23 is inconsistent between species. Patel et al suggest that CDHR1a (formerly PCDH21) fulfills the role of CDH23 in the retina.

The experiments are mainly performed using the zebrafish model system. Expression of *Pcdh15b* and *Cdhr1a* protein is shown in the photoreceptor layer through standard confocal and structured illumination microscopy. The two proteins co-IP and can induce aggregation in vitro. Loss of either *Cdhr1a* or *Pcdh15*, or both, results in degeneration of photoreceptor outer segments over time, with cones affected primarily.

The idea of the study is logical given the photoreceptor diseases caused by mutations in either gene, the comparisons to stereocilia tip links, and the protein localization near the outer segments. The work here demonstrates that the two proteins interact in vitro and are both required for ongoing outer segment maintenance. The major novelty for this paper would be the demonstration that *Pcdh15* localized to calyceal processes interacts with *Cdhr1a* on the outer segment, thereby connecting the two structures. Unfortunately, the data as presented are inadequate proof of this model.

Strengths:

The in vitro data to support the ability of Pcdh15b and Cdhr1a to bind is well done. The use of pcdh15b and cdhr1a single and double mutants is also a strength of the study, especially being that this would be the first characterization of a zebrafish cdhr1a mutant.

This is a large body of data.

Weaknesses:

- (1) I have serious concerns about the quality of the imaging here. The premise that cdhr1a/pcdh15 juxtaposition is evidence for the two proteins mediating the connection between outer segments and calyceal processes requires very careful microscopy. The SIM images have two major issues - one being that the red and green channels are misaligned and the other being evidence of bleed through between the channels. This is obvious in Fig 2A but likely true across all the panels in Fig 2, and possibly applies to confocal images in Fig 1 as well. The co-labelling with actin shows very uneven, punctate staining for actin bundles.
- (2) The newly added TEM and transverse sections include colored regions that obscure the imaging.
- (3) The quantification should be done with averages from individual fish. Counting individual measurements as single data points artificially inflates the significance. Also, the cone subtypes are still lumped together for analysis despite their variable sizes.
- (4) I highlighted previously that the measurement of calyceal processes was incorrect. The redrawn labels in Fig 7 are now more accurate, although still difficult to interpret. However, the quantification in Fig 7O is exactly the same. How can that be if the measurement region is now different?
- (5) Lower magnification views would provide context for the TEM data.
- (6) The statement describing the separation between calyceal processes and the outer segment in the mutants is still not backed up by the data.
- (7) The authors state "from the fact that rod CPs are inherently much smaller than cone CPs". This is now referenced, but incorrectly. Also, the issue of pigment interference was not addressed.
- (8) The images in panels B-F of the Supplemental Figure are uncannily similar, possibly even of the same fish at different focal planes.

<https://doi.org/10.7554/eLife.102258.2.sa1>

Author Response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Mutations in CDHR1, the human gene encoding an atypical cadherin-related protein expressed in photoreceptors, are thought to cause cone-rod dystrophy (CRD). However, the pathogenesis leading to this disease is unknown. Previous work has led to the hypothesis that CDHR1 is part of a cadherin-based junction that facilitates the development of new membranous discs at the base of the photoreceptor outer segments, without which photoreceptors malfunction and ultimately degenerate. CDHR1 is hypothesized to bind to a transmembrane partner to accomplish this function, but the putative partner protein has yet to be identified.

*The manuscript by Patel et al. makes an important contribution toward improving our understanding of the cellular and molecular basis of CDHR1-associated CRD. Using gene editing, they generate a loss of function mutation in the zebrafish *cdhr1a* gene, an ortholog of human CDHR1, and show that this novel mutant model has a retinal dystrophy phenotype, specifically related to defective growth and organization of photoreceptor outer segments (OS) and calyceal processes (CP). This phenotype seems to be progressive with age. Importantly, Patel et al. present intriguing evidence that *pcdh15b*, also known for causing retinal dystrophy in previous *Xenopus* and zebrafish loss of function studies, is the putative *cdhr1a* partner protein mediating the function of the junctional complex that regulates photoreceptor OS growth and stability.*

This research is significant in that it:

- (1) Provides evidence for a progressive, dystrophic photoreceptor phenotype in the *cdhr1a* mutant and, therefore, effectively models human CRD; and*
- (2) Identifies *pcdh15b* as the putative, and long sought after, binding partner for *cdhr1a*, further supporting the theory of a cadherin-based junction complex that facilitates OS disc biogenesis.*

Nonetheless, the study has several shortcomings in methodology, analysis, and conceptual insight, which limits its overall impact.

Below I outline several issues that the authors should address to strengthen their findings.

Major comments:

*(1) Co-localization of *cdhr1a* and *pcdh15b* proteins*

*The model proposed by the authors is that the interaction of *cdhr1a* and *pcdh15b* occurs in trans as a heterodimer. In cochlear hair cells, PCDH15 and CDHR23 are proposed to interact first as dimers in cis and then as heteromeric complexes in trans. This was not shown here for *cdhr1a* and *pcdh15b*, but it is a plausible configuration, as are single heteromeric dimers or homodimers. Regardless, this model depends on the differential compartmental expression of the *cdhr1a* and *pcdh15b* proteins. Data in Figure 1 show convincing evidence that these two proteins can, at least in some cases, be distributed along the length of photoreceptor membranes that are juxtaposed, as would be the case for OS and CP. If *pcdh15b* is predominantly expressed in CPs, whereas *cdhr1a* is predominantly expressed in OS, then this should be confirmed with actin double labeling with *cdhr1a* and *pcdh15b* since the apicobasal oriented (vertical) CPs would express actin in this same orientation but not in the OS. This would help to clarify whether *cdhr1a* and *pcdh15b* can be trafficked to both OS and CP compartments or whether they are mutually exclusive.*

First let me thank the reviewer for taking the time to comprehensively evaluate our work and provide constructive criticism which will improve the quality of our final version.

To address this issue, we are completed imaging of actin/*cdhr1a* and actin/*pcdh15b* using SIM in both transverse and axial sections (Fig 1C-H). Additionally, we have recently established an immuno-gold-TEM protocol and showcase co-labeling of *cdhr1a* and *pcdh15b* at TEM resolution along the CP (Fig 1I).

Photoreceptor heterogeneity goes beyond the cone versus rod subtypes discussed here and it is known that in zebrafish, CP morphology is distinct in different cone subtypes as well as cone versus rod. It would be important to know which specific photoreceptor subtypes are shown in zebrafish (Figures 1A-C) and the non-fish species depicted in

Figures 1E-L. Also, a larger field of view of the staining patterns for Figures 1E-L would be a helpful comparison (could be added as a supplementary figure).

The revised manuscript includes labels for the location of different cone subtypes in figure 1. All of the images showcasing CHDR1 localization across species concentrate on the PNA positive R/G cones. Larger fields of view were not collected as we prioritized the highest resolution possible and therefore collected small fields of view.

(2) Cdhr1a function in cell culture

The authors should explain the multiple bands in the anti-FLAG blots. Also, it would be interesting to confirm that the cdhr1a D173 mutant prevents the IP interaction with pcdh15b as well as the additive effects in aggregate assays of Figure 2.

The multiple bands on the WB is like our previous results (Piedade 2020), which we believe arise due to ubiquitination and proteolytic cleavage of cdhr1a. We expect the D173 mutation to result in a complete absence of cdhr1a polypeptide, based on the lack of in situ signal in our WISH studies.

Is it possible that the cultured cells undergo proliferation in the aggregation assays shown in Figure 2? Cells might differentially proliferate as clusters form in rotating cultures. A simple assay for cell proliferation under the different transfection conditions showing no differences would address this issue and lend further support to the proposed specific changes to cell adhesion as a readout of this assay.

This is a possibility; however we did not use rotating cultures, this was a monolayer culture. We did not observe any differences in total cell number between the differing transfections. As such, we do not feel proliferation explains the aggregation of K562 cells.

Also, the authors report that the number of clusters was normalized to the field of view, but this was not defined. Were the n values different fields of view from one transfection experiment, or were they different fields of view from separate transfection experiments? More details and clarification are needed.

This will be clarified in the revised manuscript, in short we replicated this experiment 3 times, quantifying 5 different fields of view in each replicate.

(3) Methodological issues in quantification and statistical analyses

Were all the OS and CP lengths counted in the observation region or just a sample within the region? If the latter, what were the sampling criteria? For CPs, it seems that the length was an average estimate based on all CPs observed surrounding one cone or one-rod cell. Is this correct? Again, if sampled, how was this implemented? In Fig 4M', the cdhr1a-/- ROS mostly looks curvilinear. Did the measurements account for this, or were they straight linear dimension measurements from base to tip of the OS as depicted in Fig 5A-E? A clearer explanation of the OS and CP length quantification methodology is required.

The revised manuscript will clearly outline measurement methods. In short, we measured every CP/OS in the imaged regions. We did not average CPs/cell, we simply included all CP measurements in our analysis. All our CP measurements (actin or cdhr1a or pcdh15), were measured in the presence of a counter stain, WGA, prph2, gnb1 or PNA to ensure proper measurements (landmark) and association with proper cell type. Our new figure 7 now includes cone OS counter staining to better highlight the OS.

All measurements were taken as best as possible to reflect a straight linear dimension for consistency.

How were cone and rod photoreceptor cell counts performed? The legend in Figure 4 states that they again counted cells in the observation region, but no details were provided. For example, were cones and rods counted as an absolute number of cells in the observation region (e.g., number of cones per defined area) or relative to total (DAPI+) cell nuclei in the region? Changes in cell density in the mutant (smaller eye or thinner ONL) might affect this quantification so it would be important to know how cell quantification was normalized.

The revised manuscript will clearly outline measurement methods. In short, rod and cone cell counts were based on the number of outer segments that were observed in the imaging region and previously measured for length. We did not observe any eye size differences in our mutant fish.

*In Figure 6I, K, measuring the length of the signal seems problematic. The dimension of staining is not always in the apicobasal (vertical) orientation. It might be more accurate to measure the *cdhr1a* expression domain relative to the OS (since the length of the OS is already reduced in the mutants). Another possible approach could be to measure the intensity of *cdhr1* staining relative to the intensity within a *Prph2* expression domain in each group. The authors should provide complementary evidence to support their conclusion.*

The revised manuscript will clearly outline measurement methods. In short, all of our CP measurements (actin or *cdhr1a* or *pcdh15*), were done in the presence of a counter stain, WGA, *prph2*, *gnb1* or PNA to ensure proper measurements and association with proper cell type.

A better description of the statistical methodology is required. For example, the authors state that "each of the data points has an n of 5+ individuals." This is confusing and could indicate that in Figure 4F alone there were ~5000 individuals assayed (~100 data points per treatment group x n=5 individuals per data point x 10 treatment groups). I don't think that is what the authors intended. It would be clearer if the authors stated how many OS, CP, or cells were counted in their observation region averaged per individual and then provided the n value of individuals used per treatment group (controls and mutants), on which the statistical analyses should be based.

This has been addressed in the revised manuscript. In short, we had an n=5 (individual fish) analyzed for each genotype/time point.

*There are hundreds of data points in the separate treatment groups shown in several of the graphs. It would not be correct to perform the ANOVA on the separate OS or CP length measurements alone as this will bias the estimates since they are not all independent samples. For example, in Figure 6H, *5dpf pcdh15b+/-* have shorter CPs compared to WT but *pcdh15b-/-* have longer compared to WT. This could be an artifact of the analysis. Moreover, the authors should clarify in the Methods section which ANOVA post hoc tests were used to control for multiple pairwise comparisons.*

We have re-analyzed the data using multiple pairwise comparison ANOVA with post hoc tests (Tukey test). This new analysis did not significantly alter the statistical significance outcome of the study.

(4) Cdhr1a function in photoreceptors

The Cdhr1a IHC staining in 5dpf WT larvae in Figure 3E appears different from the cdhr1a IHC staining in 5dpf WT larvae in Figure 1A or Figure 6I. Perhaps this is just the choice of image. Can the authors comment or provide a more representative image?

The image in figure 3E was captured using a previous non antigen retrieval protocol which limits the resolution of the cdhr1a signal along the CP. In the revised manuscript we have included an image that better represents cdhr1a staining in the WT and mutant.

The authors show that pcdh15b localization after 5dpf mirrored the disorganization of the CP observed with actin staining. They also show in Figure 5O that at 180dpf, very little pcdh15b signal remains. They suggest based on this data that total degradation of CPs has occurred in the cdhr1a^{-/-} photoreceptors by this time. However, although reduced in length, COS and cone CPs are still present at 180dpf (Figure 5E, E'). Thus, contrary to the authors' general conclusion, it is possible that the localization, trafficking, and/or turnover of pcdh15b is maintained through a cdhr1a-dependent mechanism, irrespective of the degree to which CPs are maintained. The experiments presented here do not clearly distinguish between a requirement for maintenance of localization versus a secondary loss of localization due to defective CPs.

We agree, this point has been addressed in our revised manuscript. Additionally, we have also included data from 1 and 2 year old samples.

(5) Conceptual insights

The authors claim that cdhr1a and pcdh15b double mutants have synergistic OS and CP phenotypes. I think this interpretation should be revisited.

First, assuming the model of cdhr1a-pcdh15b interaction in trans is correct, the authors have not adequately explained the logic of why disrupting one side of this interaction in a single mutant would not give the same severity of phenotype as disrupting both sides of this interaction in a double mutant.

Second, and perhaps more critically, at 10dpf the OS and CP lengths in cdhr1a^{-/-} mutants (Figure 7J, T) are significantly increased compared to WT. In contrast, there are no significant differences in these measurements in the pcdh15b^{-/-} mutants. Yet in double homozygous mutants, there is a significant reduction of ~50% in these measurements compared to WT. A synergistic phenotype would imply that each mutant causes a change in the same direction and that the magnitude of this change is beyond additive in the double mutants (but still in the same direction). Instead, I would argue that the data presented in Figure 7 suggest that there might be a functionally antagonistic interaction between cdhr1a and pcdh15b with respect to OS and CP growth at 10dpf.

If these proteins physically interacted in vivo, it would appear that the interaction is complex and that this interaction underlies both OS growth-promoting and growth-restraining (stabilizing) mechanisms working in concert. Perhaps separate homodimers or heterodimers subserve distinct CP-OS functional interactions. This might explain the age-dependent differences in mutant CP and OS length phenotypes if these mechanisms are temporally dynamic or exhibit distinct OS growth versus maintenance phases. Regardless of my speculations, the model presented by the authors appears to be too simplistic to explain the data.

We agree with the reviewer, as such we have revised the discussion in our revised manuscript.

Reviewer #2 (Public review):

Summary:

The goal of this study was to develop a model for CDHR1-based Con-rod dystrophy and study the role of this cadherin in cone photoreceptors. Using genetic manipulation, a cell

*binding assay, and high-resolution microscopy the authors find that like rods, cones localize CDHR1 to the lateral edge of outer segment (OS) discs and closely oppose PCDH15b which is known to localize to calyceal processes (CPs). Ectopic expression of CDHR1 and PCDH15b in K652 cells indicates these cadherins promote cell aggregation as heterophilic interactants, but not through homophilic binding. This data suggests a model where CDHR1 and PCDH15b link OS and CPs and potentially stabilize cone photoreceptor structure. Mutation analysis of each cadherin results in cone structural defects at late larval stages. While *pcdh15b* homozygous mutants are lethal, *cdhr1* mutants are viable and subsequently show photoreceptor degeneration by 3-6 months.*

Strengths:

A major strength of this research is the development of an animal model to study the cone-specific phenotypes associated with CDHR1-based CRD. The data supporting CDHR1 (OS) and PCDH15 (CP) binding is also a strength, although this interaction could be better characterized in future studies. The quality of the high-resolution imaging (at the light and EM levels) is outstanding. In general, the results support the conclusions of the authors.

Weaknesses:

While the cellular phenotyping is strong, the functional consequences of CDHR1 disruption are not addressed. While this is not the focus of the investigation, such analysis would raise the impact of the study overall. This is particularly important given some of the small changes observed in OS and CP structure. While statistically significant, are the subtle changes biologically significant? Examples include cone OS length (Figures 4F, 6E) as well as other morphometric data (Figure 7I in particular). Related, for quantitative data and analysis throughout the manuscript, more information regarding the number of fish/eyes analyzed as well as cells per sample would provide confidence in the rigor. The authors should also note whether the analysis was done in an automated and/or masked manner.

First let me thank the reviewer for taking the time to comprehensively evaluate our work and provide constructive criticism which will improve the quality of our final version.

The revised manuscript outlines both methods and statistics used for quantitation of our data. (please see comments from reviewer 1). While we do not include direct evidence of the mechanism of CDHR1 function, we do propose that its role is important in anchoring the CP and the OS, particularly in the cones, while in rods it may serve to regulate the release of newly formed disks (as previously proposed in mice). We do plan to test both of these hypothesis directly, however, that will be the basis of our future studies.

Reviewer #3 (Public review):

Summary:

The manuscript by Patel et al investigates the hypothesis that CDHR1a on photoreceptor outer segments is the binding partner for PCDH15 on the calyceal processes, and the absence of either adhesion molecule results in separation between the two structures, eventually leading to degeneration. PCDH15 mutations cause Usher syndrome, a disease of combined hearing and vision loss. In the ear, PCDH15 binds CDH23 to form tip links between stereocilia. The vision loss is less understood. Previous work suggested PCDH15 is localized to the calyceal processes, but the expression of CDH23 is inconsistent between species. Patel et al suggest that CDHR1a (formerly PCDH21) fulfills the role of CDH23 in the retina.

The experiments are mainly performed using the zebrafish model system. Expression of Pcdh15b and Cdhr1a protein is shown in the photoreceptor layer through standard confocal and structured illumination microscopy. The two proteins co-IP and can induce aggregation in vitro. Loss of either Cdhr1a or Pcdh15, or both, results in degeneration of photoreceptor outer segments over time, with cones affected primarily.

The idea of the study is logical given the photoreceptor diseases caused by mutations in either gene, the comparisons to stereocilia tip links, and the protein localization near the outer segments. The work here demonstrates that the two proteins interact in vitro and are both required for ongoing outer segment maintenance. The major novelty of this paper would be the demonstration that Pcdh15 localized to calyceal processes interacts with Cdhr1a on the outer segment, thereby connecting the two structures. Unfortunately, the data presented are inadequate proof of this model.

Strengths:

The in vitro data to support the ability of Pcdh15b and Cdhr1a to bind is well done. The use of pcdh15b and cdhr1a single and double mutants is also a strength of the study, especially being that this would be the first characterization of a zebrafish cdhr1a mutant.

Weaknesses:

(1) The imaging data in Figure 1 is insufficient to show the specific localization of Pcdh15 to calyceal processes or Cdhr1a to the outer segment membrane. The addition of actin co-labelling with Pcdh15/Cdhr1a would be a good start, as would axial sections. The division into rod and cone-specific imaging panels is confusing because the two cell types are in close physical proximity at 5 dpf, but the cone Cdhr1a expression is somehow missing in the rod images. The SIM data appear to be disrupted by chromatic aberration but also have no context. In the zebrafish image, the lines of Pcdh15/Cdhr1a expression would be 40-50 um in length if the scale bar is correct, which is much longer than the outer segments at this stage and therefore hard to explain.

First let me thank the reviewer for taking the time to comprehensively evaluate our work and provide constructive criticism which will improve the quality of our final version.

To address this issue, we have added images of actin/cdhr1a and actin/pcdh15b using SIM in both transverse and axial sections. Additionally, we have established an immuno-gold-TEM protocol and provide data showcasing co-labeling of cdhr1a and pcdh15b at TEM resolution.

(2) Figure 3E staining of Cdhr1a looks very different from the staining in Figure 1. It is unclear what the authors are proposing as to the localization of Cdhr1a. In the lab's previous paper, they describe Cdhr1a as being associated with the connecting cilium and nascent OS discs, and fail to address how that reconciles with the new model of mediating CP-OS interaction. And whether Cdhr1a localizes to discrete domains on the disc edges, where it interacts with Pcdh15 on individual calyceal processes.

The image in figure 3E was captured using a previous non antigen retrieval protocol which limits the resolution of the cdhr1a signal along the CP. In the revised manuscript we include an image that better represents cdhr1a staining in the WT and mutant.

(3) The authors state "In PRCs, Pcdh15 has been unequivocally shown to be localized in the CPs". However, the immunostaining here does not match the pattern seen in the Miles et al 2021 paper, which used a different antibody. Both showed loss of staining in pcdh15b mutants so unclear how to reconcile the two patterns.

We agree that our staining appears different, but we attribute this to our antigen retrieval protocol which differed from the Miles et al paper. We also point to the fact that pcdh15b localization has been shown to be similar to our images in other species (monkey and frog). As such, we believe our protocol reveals the proper localization pattern which might be lost/hampered in the procedure used in Miles et al 2021.

(4) The explanation for the CRISPR targets for cdhr1a and the diagram in Figure 3 does not fit with crRNA sequences or the mutation as shown. The mutation spans from the latter part of exon 5 to the initial portion of exon 6, removing intron 5-6. It should nevertheless be a frameshift mutation but requires proper documentation.

This was an overlooked error in figure making, we have corrected this typo in the revised manuscript.

(5) There are complications with the quantification of data. First, the number of fish analyzed for each experiment is not provided, nor is the justification for performing statistics on individual cell measurements rather than using averages for individual fish. Second, all cone subtypes are lumped together for analysis despite their variable sizes. Third, t-tests are inappropriately used for post-hoc analysis of ANOVA calculations.

As we discussed for reviewer 1 and 2, all methods and quantification/statistics will be clearly described in the revised manuscript.

(6) Unclear how calyceal process length is being measured. The cone measurements are shown as starting at the external limiting membrane, which is not equivalent to the origin of calyceal processes, and it is uncertain what defines the apical limit given the multiple subtypes of cones. In Figure 5, the lines demonstrating the measurements seem inconsistently placed.

As we discussed for reviewer 1 and 2, all methods and quantification/statistics will be clearly described in the revised manuscript. We have also clarified that CP measurements were made based on a counterstain for the cone/rod OS so that the actin signal was only CP associated. We have included the counter stain in our revised Figure 7.

(7) The number of fish analyzed by TEM and the prevalence of the phenotype across cells are not provided. A lower magnification view would provide context. Also, the authors should explain whether or not overgrowth of basal discs was observed, as seen previously in cdhr1-null frogs (Carr et al., 2021).

The revised manuscript now includes the n number for our TEM samples. We have also added text comparing our results directly to Carr 2021.

(8) The statement describing the separation between calyceal processes and the outer segment in the mutants is not backed up by the data. TEM or co-labelling of the structures in SIM could be done to provide evidence.

We have completed both more SIM as well as immuno-gold TEM to support our conclusions, see new Figure 1.

(9) "Based on work in the murine model and our own observations of rod CPs, we hypothesize that zebrafish rod CPs only extend along the newly forming OS discs and do not provide structural support to the ROS." Unclear how murine work would support that conclusion given the lack of CPs in mice, or what data in the manuscript supports this conclusion.

In the revised manuscript we have adjusted our discussion to hypothesize that the small length of rod CPs is most likely to represent their interaction with newly forming discs rather

than connect with mature discs which are enclosed in the OS.

(10) The authors state "from the fact that rod CPs are inherently much smaller than cone CPs" without providing a reference. In the manuscript, the measurements do show rod CPs to be shorter, but there are errors in the cone measurements, and it is possible that the RPE pigment is interfering with the rod measurements.

We have included references where rod CPs have been found to be shorter. We have no doubt that in zebrafish the rod CPs are significantly shorter. All our CP measurements are done with a counter stain for rods and cones to be sure that we are measuring the correct cell type.

*(11) The discussion should include a better comparison of the results with ocular phenotypes in previously generated *pcdh15* and *cdhr1* mutant animals.*

The revised manuscript has included these points.

(12) The images in panels B-F of the Supplemental Figure are uncannily similar, possibly even of the same fish at different focal planes.

We assure the reviewer that each of the images in supplemental figure 1 are distinct and represent different in situ experiments.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) In the second sentence of the Introduction section, the acronym 'PRC' should be defined.

This has been corrected

*(2) In the Discussion section, it would be useful to comment on differences between the published *Xenopus cdhr1*^{-/-} OS phenotypes and the published zebrafish *pcdh15b*^{-/-} OS phenotypes compared to the present zebrafish *cdhr1a*^{-/-} phenotypes. In the published studies, OS in these mutants demonstrated dysmorphic and overgrown disc membranes compared to the relatively minor disc layering defects shown for *cdhr1a*^{-/-} in the present study.*

This discussion has been added.

(3) CDHR1 mutations in patients cause cone-rod dystrophy, but mutations in PCDH15 (Usher 1F) cause rod-cone dystrophy. In the Discussion section, the authors should comment on what might lead to these different phenotypic trajectories in humans in the context of their proposed model.

We have added to our discussion highlighting that is not possible to assess rod-cone dystrophy in the *pcdh15b* model as the mutation is lethal by 15dpf, which is still before most rods mature.

Reviewer #2 (Recommendations for the authors):

In addition to defining the 'n' for animal and cell numbers (as well as methods of analysis - automated/masked), there are a few additional recommendations for the authors.

(1) Expression of USH1 genes in larval zebrafish (Figure S1) is not very convincing. SC RNAseq data exists and argues against this cell type restriction.

Based on extensive experience with WISH we are confident that our interpretation of the data are valid. Furthermore, analysis of the daniocell data base confirms that *cdh23*, *ush1ga*, *ush1c* (harmonin) and *myo7aa* all have either no expression in photoreceptors or very low levels especially compared to *pcdh15b* and *cdhr1a*.

(2) The model in Figure 1 is great. The coloring was a bit confusing. *Cdhr1* and axoneme are both in green, while *Pcdh15* and actin are both in red. Can each have its own color?

Changed *pcdh15b* color to blue

(3) Figure 2A: Please explain the multiple bands in some lanes. What do the full blots look like?

Full blots were uploaded to eLife and do not exhibit any additional bands. The multiple bands are likely due to ubiquitination or proteolytic cleavage of *cdhr1a* and have been documented in our previous publication (Piedade 2020).

(4) Is "data not shown" permissible? (lack of compensation of *cdh1b* in *cdh1a* mutants) (nonsense-mediated decay of the mutant transcript).

We have added a supplementary figure showcasing this data.

(5) Figure 4: Is there a TEM phenotype in discs before 15dpf? One would think there would be...?

Due to technical limitations, we have not been able to examine disc phenotypes prior to 15dpf.

(6) Figure 5: How are calyceal processes discriminated from cortical/PM-associated actin? A bonafide calyceal marker seems to be needed. *Espin* or *Myo3*, for example.

We discriminate to identify CPs as actin signal that originates at the base of the OS and travels along the OS. *Pcdh15b* is a bonafide CP marker which we show overlaps with actin signal along CPs.

(7) Figures 5A-J: How is actin staining for CPs discriminating between rod and cones??? Apical - basal level imaging? This could be better clarified.

CP identification is based on co-stain for either rod or cone Oss

(8) Figure 6: Het phenotype for *pcdh15b*+/- (cone OS length and CP length at 5 and 10 dpf) is surprising ... worth discussing. (Figures 6E, H).

The discussion section has been updated to discuss this finding.

(9) Last, the authors state "Data not shown" throughout the manuscript. I do not believe this is allowed for the journal.

This data (*cdhr1b* expression in *cdhr1a* mutants as well as *cdhr1a* WISH in *cdhr1a* mutants) has been added as supplementary figures.

Reviewer #3 (Recommendations for the authors):

Major comments are addressed above and the most important is the need for a convincing demonstration of *Cdhr1a* localization on the outer segment and proximity to *Pcdh15b*. The SIM could be a powerful tool, but the images provided are impossible to assess without any basis for context. Could a membrane, *Prph2*, and/or actin label be added? And lower magnification views?

Minor comments.

(1) The mention of "short CPs" in rodents is not an accurate description. Particular rodents (e.g. mouse, rat) lack CPs altogether or have a single vestigial structure.

We have adjusted the text to reflect this point.

(2) Inconsistent spacing between numbers and units.

We have corrected these inconsistencies

(3) Missing references.

We have added missing references

(4) Indicate the mean or median for bar graphs.

The materials and methods section now specifies that all of our graphs depict a mean value

(5) Unclear how rods are distinguished from cones in the cone analysis if both are labeled with prph2 antibody.

Rods are physiological separate from cones in zebrafish retina and therefore easily identified by location as well as their distinct pattern of actin staining.

(6) Red and green should not be used together for microscopy images.

(7) The diagram in Figure 1D is confusing because of the repeated use of red and green for disparate structures. Also, the location and structure of actin are misrepresented, as is the transition of disc structure during maturation in rods.

We have adjusted the color of pcdh15b to blue.

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